The Curious Incident of the Fork in the Night-Time

Number Basher

Based on *The Curious Incident of the Dog in the Night-Time*

Chapter 2

12:07:57 AM. A house. I know it had belonged to Mrs. Shears. A lawn. In the middle, a dog, lying on the grass. Body laid down, as if running. Eyes closed, as if sleeping. But the dog was neither running nor asleep.

The dog was dead.

A garden fork, sticking out of the dog. It was slanted yet had not fallen over. It must have impaled the dog onto the ground. So the dog was presumably killed by the fork, unless the garden fork was stuck into the dog after he had died.

There appeared to be no wounds other than the bloody stain surrounding the fork, but there could be wounds on the side of the dog in contact with the grass, where some blood is leaking out slowly, but that would also be logically explained if the fork had indeed penetrated the entirety of the body of the dog. There may also be internal wounds, which I would not be able to see easily.

So I made a *Fermi calculation*. There is a less than 0.1% probability that the dog died due to an outside cause, and a further less than 0.01% probability that a fork would be stuck into him afterwards dependent on his death, and a less than 30% chance that no detectable traces would be left. But there is a greater than 0.01% probability that a fork will be used to kill a dog. If one were to kill a dog with the tools nearby, only a fork would work well. (The drill may also be used, but that would create undesirable noise.)

The dog was hence killed with the fork with probability over 99.96%.

I could be certain about this.

I went in. Standing beside the dog, I knelt, and touched him. The muzzle of the dog was still warm. So he was alive no more than an hour ago, and the lower bound on the probability that a fork had killed him had rose up to 99.99%.

So, for all intents and purposes, the fork had killed him, unless evidence shows otherwise.

The dog was called Wellington. He belonged to Mrs. Shears, who is our friend. She lived in an isolated house in the middle of the neighborhood.

I stroked Wellington and wondered what had killed him, and why.

Chapter 3

I have not introduced myself. My name is Christopher. I have a photographic memory, which many scientists believe is impossible or not scientifically grounded. I memorised all the countries of the world, their capital cities, and population. I was going to memorise the prime numbers until I realised that basic primality testing (using the Miller–Rabin method on 2 and 3) worked for numbers up to a million and that according to the prime number theorem there would be about 70,000 to 80,000 prime numbers up to a million. I would easily win *Who Wants to Be a Millionaire*, but I have never been given the chance. I will probably tell you more about myself when I have more time. (The word *probably* is added here for plausible deniability.)

I am introducing myself because I am the main character of this story. I want to figure out what had killed Wellington, and *when there is a will there is a way.*

I pulled the fork out of the dog, lifted him into my arms, and hugged him.

I like dogs.

I am a *dog-person*, which means I am emotionally attached to dogs. Siobhan says that this is because animals tend to be more predictable and truthful. I agree — at least the predictability in them makes them much easier to deal with humans. However, people I work with seem to be very honest. Some teachers like to think that they can keep secrets from students, but these plans always backfire because the students end up discovering them anyways. *I hate it when that happens.*

It was 12:13:14 AM when I heard screaming. I looked up and saw Mrs. Shears running towards me from the patio.

"What in fuck's name have you done to my dog?"

I do not like it when people shout at me. Swearing in particular makes me scared that they are going to hit me or touch me and I do not know what is going to happen.

I also do not like it when people touch me.

"Let go of the dog."

"Let go of the fucking dog for Christ's sake."

I was scared, so I put the dog down on the lawn and moved back three steps. She bent down to inspect the situation, and I thought she was going to pick the dog up herself, but she didn't, perhaps because she noticed how much blood there was and didn't want to get dirty. She started screaming again.

I put my hands over my ears and rolled forward onto the grass because the grass was wet and cold. It was nice.

Chapter 5

I realise that I have mentioned Siobhan without introducing her. I am very sorry.

Ten years ago, I was diagnosed with autism, level 2. I still *remember it like it was yesterday*, because my memory goes all the way to when I was four.

Many people say being autistic (I prefer identity-first language) is a gift, and I agree. Mathematics is my *special interest*, and I am very talented in it. I am also very exercised in rationality, and combined with my perfect memory and calculation speed, I can do many things. However, identifying emotions is not one of these things.

When I was seven, I was assigned to Siobhan, a paraprofessional. When I first met Siobhan, she showed me a picture like this



which meant *sad*, like what I felt when I found the dead dog.

Then she showed me this picture



but I did not know what that meant since it was too cryptic and I could not handle real human faces, so I inferred from the flower in the background that it meant *happy* like when I am still awake at three or four in the morning and was all alone, walking in the empty street. Siobhan said that she had pulled an image on the internet and added a grey overlay, hoping that that would eliminate background information, and that in real world, there would not be flowers when someone is happy.

Hence, I got Siobhan to find lots of these faces and I tried to find all the patterns, which is why I am a lot better with faces now.

Some people say that being autistic is just *taking a different perspective on the world*, but *there is more to it*, and when people say *autism is not a disability* they nullify the obstacles we encounter in life, and that is *sad*.

Chapter 7

Then the police arrived. I do not like the police. Their appearance is *Bayesian Evidence* for *something is horribly wrong.*

The policewoman took Mrs. Shears back toward the house. I lifted my head off the grass, now that the screaming was gone.

"Would you like to tell me what's going on here?"

"The dog is dead."

"I'd got that far."

"The dog is probably killed by a fork."

"How old are you?"

"I am 15 years old."

"And what, precisely, were you doing in the garden?"

"I was holding the dog."

"And why were you holding the dog?"

"I like to hold dogs."

"Did you kill the dog?"

"No."

"Is this your fork?"

"No."

"You seem very upset about this."

"Yes."

"Are you telling the truth?"

"Yes."

He was asking too many questions too quickly and my brain was having trouble evaluating all the options properly so I was turning on default mode and just presented whichever was the closest binary answer without considering the consequences.

"I am going to ask you once again..."

And suddenly a ray of red light entered my eyes, originating from the police car, and that had served to be *the final straw*, so I *broke down* against the grass.

The policeman took hold of my arm and lifted me onto my feet.

My brain does not like it when people grab me.

And this is when I hit him.

Chapter 11

This is a serious book.
I like puns and wordplay. Here is one that I like:

> His face was drawn but the curtains were real.

When I imagine this pun, I think of two *parallel universes* like so:



I like puns because I think they are cleverly constructed. A single word has the power to create an ambiguity forking into two contrasting realms. *Isn't that fantastic?*
I also like cryptic clues. One clue that I constructed was *cast game (9)*:



But this book is a serious book, so do not expect puns here.

Chapter 13

I did not mean to hit the policeman. I certainly did not mean to hurt the policeman. My brain just wanted to get away from him. Touching makes me feel uncomfortable. Siobhan had told me that this kind of reaction was reflexive, so I do not have to be guilty. But I still feel guilty because I punched a police officer and that was a wrong thing to do.

The policeman looked at me for a while without speaking.

"I am arresting you for assaulting a police officer."

"Yes."

I had recovered and so I oriented myself against the direction of the light and I was now a lot calmer.

"I strongly advise you to get into the back of the police car, because if you try any of that monkey business again, you little shit, I will seriously lose my rag. Is that understood?"

"Yes."

I walked over to the police car, which was parked just outside the gate.

"The little bugger just had a pop at me, Kate. Can you hang on with Mrs. S. while I drop him off at the station? I'll get Tony to swing by and pick you up."

"Sure. I'll catch you later."

"Okey doke."

The scenery was nice, but I had no time to enjoy it. I had to think. Who killed Wellington?

Given the circumstances, one can almost be sure that the killing was deliberate and not accidental. It is improbable to *just so happen* to hit the dog throwing the fork, and there was no reason to throw the fork anyways. The fork had to be thrown very hard for it to have impaled the dog completely. The fork could also have been intentionally planted in the ground without knowledge of the dog, but the angle of the fork was weird for it to have been planted down, since otherwise it would have been straight down. So the killing was likely deliberate.

In that case, we have to have a motive, but I did not have anything in mind.

And then we were at the police station.

Chapter 17

This is a murder mystery novel. I like novels. They are fun and exciting, and they give you a literal *other-worldly experience*.

But I dislike it when the descriptions become metaphorical and the intended meanings are hidden behind obfuscation. In *The Waves*, Virginia Woolf writes "Veined as I am with iron, with silver and streaks of common mud, I cannot contract into the firm fist which those clench who do not depend on stimulus." The excerpt is supposed to be about Bernard's dependence on external forces to preserve his determination, but it took me a long time to understand it since I didn't know why iron, silver, and mud was there at that time and only later knew that they were intended to convey strength and toughness in a metaphorical way. (I also skipped over a bit and didn't know what the "firm fist" referred to. That was the last time I attempted to skip parts of a novel.)

I especially like murder mystery novels, so I am writing a murder mystery novel.

A murder mystery novel is a puzzle, and if it is fair like a *Fair-Play Whodunnit* then the reader is supposed to solve the puzzle before the solution is revealed. Siobhan read the first chapter of my novel and said that it was "different" in one of those contexts where quotation implies a stretch in naming. She said that it was because usually it is people who were killed in murder mystery novels. She said that this was because people were more important than dogs but I tried to interrupt and argue how rationalists haven't agreed on their relative importance and some dogs can be more intelligent than some humans like Steve who has Down Syndrome and she just said how it is really that people cared more about other people and not dogs.

So I explained why I had to write about something real since I couldn't make things up in a consistent manner. I do not tell lies. Mother used to say that this was because I was a good person. I believe I am a good person, but correlation does not imply causality. I don't tell lies because I can't tell lies.

A lie is when you say something that is objectively false. Even though we cannot exactly predict the future due to the *Butterfly Effect* and the *Uncertainty Principle*, some scenarios are provably inconsistent and *Gödel's Incompleteness Theorem* demonstrates that consistency is undecidable. And so it is very difficult to maintain the consistency of a lie with lies.

And this is why everything I have written here is true to the best of my knowledge.

Chapter 19

Chapters in books are usually given the cardinal numbers 1, 2, 3, 4, 5, 6 and so on. But I have decided to give my chapters the prime numbers 2, 3, 5, 7, 11, 13 and so on because I like prime numbers. Pragmatically, this makes no sense. It took me very long to get this to typeset in LaTeX. It also makes navigation hard. But they mean something special to me, since they are what remains when you take off all the patterns.

This is how you work out what prime numbers are.

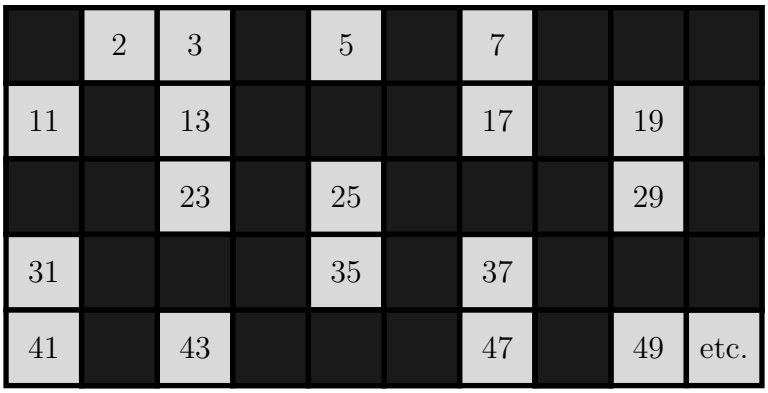
First you write down all the positive whole numbers in the world. Well, that would be impossible, since there are infinitely many of them. But you can write down however many you think is enough. We'll use 50. Take away 1.

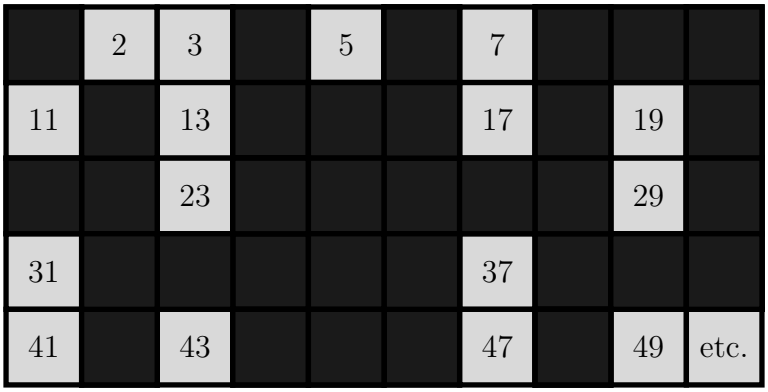
	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	etc.

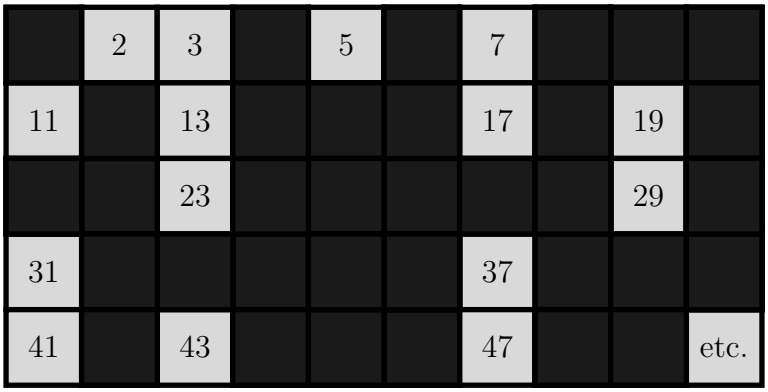
Then you look at the first remaining number, which is 2, and you remove all the multiples of 2 except itself.

	2	3		5		7		9	
11		13		15		17		19	
21		23		25		27		29	
31		33		35		37		39	
41		43		45		47		49	etc.

And then you do the same for 3, then 5, then 7. Since $\lfloor \sqrt{n} \rfloor = 7$ (where n is the number of numbers we have), we are done.







The numbers that are left are the prime numbers.

This is called the *Sieve of Eratosthenes*, and it is extremely efficient if you want all the prime numbers up to a certain n. In asymptotic notation, the algorithm is $\Theta(\sqrt{n})$, which means that if n is large enough then quadrupling n roughly doubles the runtime of the algorithm.

The rule for working out prime numbers is really simple, but no one has ever worked out a simple formula for telling you whether a very big number is a prime number.

Many people think that if a number is really, really big, it can take a computer years to work out whether it is a prime number. But that requires astronomical numbers, and what they really mean is that factoring large numbers are hard, and this is the foundation for cryptography. For instance, my computer can check that this is not a prime number within seconds:

596616896060596946589784043222157068224096249140547044249286562818564900430
507674288857135330862080717352869573008041225717284733319254801298748422623
049623846590287544602250373467602952406325626039220669076855039577218013372
431449859755200787000296914266526002155764418275765341852657863751188707364
091425184661818467827771355346874509843838601180887233587789059499937789175
785296622159141082248098505317720316478789216499682758661319005230756029562
147843410651050973879907759269019347176961345357629386940088193093479478033
587132435260394683899069468386269622015835598421975924225184124860813228 21
578885655977357509028021788302888998500670167620857808063019843665423164508
864846214967477760894188054017210602695558215469985824355928415001826614378
931820535982654149991290468522980134109816251114545547096683826369198474055
910849108831508433593651678986614224790747563823533274390923657645917370466
244603194012009241982665306603520875150029318741700035383593064774733140229
628620277737531415705993433077241155149657458033174873566031934263382903336
385949737616219492140036954391126237237257920286784271217887101827382215815
587414519788660424374444818465174002132280530984233534107161482710135808190
156678343387861184158772408565076506973360750891155045418240869110290573427
464112723167624073864370338920056992274079045239805378654992128036282225889
242355502622133707522924509594670741353445974060714369491122453023986094179
905324613739176207807095342069372050209895712648532763245391390020593207071
576313717831975352457881525716078239647573042502250655316077363904248390654
799171428247601455031341673198783577188592045358156928546996028524741830556
175003442401296217327031429125272870169407345336192732580500358572756895776
737764300176258724773914212 7

Chapter 23

When I got to the police station they made me empty my pockets. Then they took the laces out of my shoes, which was weird. I think somebody probably tried to do something with the laces, like tried to commit suicide. There are so many other things that can help one commit suicide like the shards that the pieces of glass on my glasses can break into that only taking away the shoelaces didn't make sense unless somebody tried using it before.

I took out what I had from within my pockets:

1. A Swiss Army Knife with 15 attachments
2. A piece of string
3. A Rubik's Cube
4. Three pellets of rat food for my rat, Toby
5. £2.47 with two 10p coins instead of a 20p coin
6. A credit card Father had given me
7. My house's front door key

Then, they took off my watch, and I thought that surely they would take my glasses too since glasses produce so much more vulnerability than watches, but they gave me back my glasses. I checked my watch one last time. It was 1:06:12 AM.

They asked me if I have any family. I said I do, and I inferred that they wanted to know who my family was so I told them that there were Father and Mother, but Mother was dead. And I would have went on with Uncle Terry and my Grandma but I thought that was sufficient information and so I paused slightly and they asked me for Father's phone number so I told them his two numbers and didn't talk about Uncle Terry and my Grandma.

The police cell was fine. I wanted to measure the total volume of air so I *took an estimate*. It looks to be about 1.95 meters long and wide, but 2.2 meters high, so the total area was around 8.36 cubic meters. Actually it is 8.3655, since I introduced a 0.0025 error term when approximating 1.95^2. But there was no risk of running out of oxygen since it had a small window with bars spaced 10 centimeters apart.

I wondered how I would escape if I was in a story. I decided that my best plan would be to memorise the guards schedules and wait for a sunny day to use my glasses to focus the sunlight on my clothing and start a fire when I knew they would be around and see the smoke, and I would then escape in the chaos. If they didn't notice I could always put the fire out with another clothing soaked with the water they gave us to drink. But I probably didn't need to escape since I only committed a minor offense.

Chapter 29

I find people confusing. This is because people can be very ambiguous.

People use a lot of facial expressions and body language to communicate, and yet nobody teaches these methods to people like me who don't understand them naturally. Not only that, but they are heavily dependent on context. *Raising an eyebrow* can mean anything from "that is very intriguing" to "that is very stupid".

People also make metaphors. I know a lot of metaphors like

> The dog was *stone dead*.

which isn't really about stone and stone is just used to convey the rigidness of the dog's body similar to the rigidness of the stone. But when I come across a metaphor that I cannot identify, I get confused since I sometimes don't realise that it is a metaphor.

One interesting fact about the word "metaphor" is that it means carrying something from one place to another, and it comes from $\mu\varepsilon\tau\alpha$ (which means from one place to another) and $\varphi\varepsilon\rho\varepsilon\iota\upsilon$ (which means to carry), and it is when you describe something by using a word for something that it isn't without using any indicator words, which are "like" and "as ... as". This means that the word metaphor is a metaphor when it is used metaphorically to mean assigning the attributes of a phrase to another.

I like metaphors because they are interesting, but sometimes I don't recognise them and I take them literally.

My name is a metaphor. It means carrying Christ and it comes from $\chi\rho\iota\sigma\tau o\varsigma$ (which means Jesus Christ) and $\varphi\varepsilon\rho\varepsilon\iota\upsilon$ and it was the name given to St. Christopher because he carried Jesus Christ across a river.

But these are small issues. I hate it when people tell me lies *for the greater good*. I like to assume that everything that is said to me is true. That means I get to make direct deductions instead of probabilistic inferences and that saves me a lot of work. But people lying, especially when it is to me because they think that I am naive, is really irritating.

And this is why all of my guesses are based on probability, because *anyone who tells you they're certain about how the world works is either a mathematician, or they're selling something.*

Chapter 31

I thought it was around 30 minutes later when Father arrived at the police station, but I did not see him until another two minutes later and it was 1:28:17 AM then so I was off by more than 30% since I am quite dependent on my watch.

I couldn't hear the entire conversation, but I imagined this to be a plausible completion:

"I want to see my son."

"He is locked up right now." (This explains why Father was suddenly infuriated.)

"Why the hell is he locked up?"

"Let's not be angry." (This seemed unlikely, but was one of the more pausible reasons.)

"Of course I'm fucking angry."

And so another thirty seconds later a policeman let me out. I saw his watch which said 1:28:47 and so I knew it was 1:28:17 earlier.

I stepped outside and I saw Father standing in the corridor. He held up his right hand and spread his fingers out, which is a gesture with a meaning equivalent to a hug to me since my brain doesn't like it when I am hugged. I mirrored his motion and we made our fingers and thumbs touch each other. It means that he loves me.

Then we followed the policeman down the corridor to another room with a table and three chairs and there was a tape recorder but it was powered off.

"I have spoken to your father and he says that you didn't mean to hit the policeman."

I corrected him slightly but confirmed the intent of the statement. "I only meant to get away from him. I am autistic, and I just wanted him to stop touching me. I couldn't help hitting him."

He was quiet for a few seconds. I didn't know if he had believed me, because autism is a rare disorder and it is probable that a person committing a crime would use it as defense — more probable then actually happening to have that disorder.

"Did you kill the dog, Christopher?"

Father probably told him my name over the phone.

"No."

"Do you know that it is wrong to lie to a policeman and that you can get into a very great deal of trouble if you do?"

"Yes."

"So, do you know who killed the dog?"

"No."

"Are you telling the truth?"

"Yes."

And then he said that he would give me a caution and I knew that they are going to keep a record of what I just did, and that if I get into any more trouble they will *take things much more seriously.* And I said that I understood.

Chapter 37

Mother had died 2 years ago.

I had come home from school and opened the door with my key but nobody was there.

An hour and a half later Father came home from work and asked whether I had seen Mother and I said that I hadn't. Then he made some phone calls and came up to my room and said he had to go out for a while.

That while was 2 hours, 32 minutes, and 25 seconds. When he came back, he was sitting in the kitchen staring out of the back window down the garden to the pond so I thought he was thinking about something.

"I'm afraid you won't be seeing your mother for a while."

He didn't look at me when he said this. Usually people look at you when they're talking to you, so that was weird, but he was probably just sad.

"Why not?"

He waited for 87 seconds, and then he said that my mother has had to go into hospital. That was weird as well, since it shouldn't take so long to say this objective truth. But the truth, however objective would have been hard to say.

So I asked if we could visit her, and he said no. I had known that he would say no, but my question was a transition to my next.

So I asked why we couldn't visit her, and he said that she needed rest. So I asked what happened to mother, and Father paused again, and then said that she has a problem with her heart, and that pause seemed strangely *out of place*. And then I asked if we should take her food, and Father said that he would do it. And then I asked him some more questions and he didn't want to answer them.

Father told me two weeks later that Mother was dead. He said that she had been looking OK and seemed to be getting better. He said that she died of a heart attack and it wasn't expected.

I was devastated. I'd taken her for granted, and suddenly she was gone. I don't feel now what I felt then, but I still wish she'd come back. I still dream about it, still think of the things I'd tell her if she were here. I can still be brought to tears by something that dredges up the right memories. I remember that I went out to Mrs. Shears' garden and I laid myself down onto the grass, which was also wet, and I cried and cried until my Father called me to dinner.

> (And somewhere, in the back of my mind was a small, small note of confusion, a sense of something wrong about that explanation; and it should have been a part of my art to notice that tiny note, but I was distracted. For it is a sad rule that whenever you are most in need of your art as a rationalist, that is when you are most likely to forget it.)

Since then, Mrs. Shears had cared for me instead of Mother. That day, she made us spaghetti and tomato sauce, and after dinner she played Scrabble with me and I only beat her 395 points to 134 because I just wanted to end the game and go to sleep.

Chapter 41

There were clouds in the sky on the way home, so I couldn't see the same scenery.

"I'm sorry."

Father had come for me, which had taken his time.

"It's OK."

"I didn't kill the dog."

"I know." (I have learned that other people say this without actual consideration, and so Father may or may not have actually known it but had just said it to indicate that he had heard my statement.)

"Christopher, you have to stay out of trouble, OK?"

And suddenly, I saw how utterly stupid I was.

Of course any police would have thought that I killed the dog, since I was the only one there except Mrs. Shears and Mrs. Shears didn't seem to kill the dog.

"I'm sorry. I was stupid."

But I am not very good with what other people think of my actions. When I was little, Siobhan sat down at a desk next to me and put a tube of Smarties on the desk. She asked what I thought was inside there, and I guessed Smarties.

Then she took the top off the Smarties tube and turned it upside down and a little red pencil came out, and then she put the little red pencil back inside the Smarties tube and put the top back on.

Then she asked what Mother would have said if she came in now and we asked her what was inside, and I said "a pencil" which was wrong since Mother did not see the pencil and the best guess with only her information would still be Smarties.

Later, I found out that this was called the Sally-Anne test, and that it was given to me because they suspected that I am autistic.

Now, when I think of this as a logic puzzle, I can solve it. But it makes me exhausted to think about all the different perspectives all the time, which is why I only do it retrospectively.

I knew that I had to find out who killed Wellington, but I don't have to tell Father that.

When I went back to home I turned my computer on and played 76 games of Minesweeper and got 107 seconds on the Expert Version, which was *8 seconds off* my best time, which was 99 seconds.

Chapter 43

The next day, I decided that I would try and find out who killed Wellington.

7:04:01 PM. I went round to Mrs. Shears's house and knocked on the door and waited for her to answer it.

When she opened the door she was holding a mug of tea and she was wearing sheepskin slippers and she had been watching a quiz program

> The capital city of Venezuela is...
>
> (a) Maracas
>
> (b) **Caracas**
>
> (c) Bogota
>
> (d) Geourgetown

"Christopher, I really don't think I want to see you right now."

I *took this in.*

"I have something to ask you."

I waited for a second in case she had any objections.

"Do you know who killed Wellington?"

Some of her tea spilled onto the carpet. I knew that meant that she was in shock, but I don't know what she is in shock of. So I analyzed and I realised that I was being stupid again, because just like how the police thought it was me who had killed Wellington, she would also have thought I had killed Wellington, and in that case I don't know how she would have felt when I asked this question.

"I didn't kill Wellington," I explained, "I saw him lying there and I wanted to pet him. Do you know who killed Wellington?"

She didn't answer, so I thought that perhaps I had not convinced him, and so I attempted to use *logos* and *pathos*, which meant convincing with logic and emotion.

"Mrs. Shears, you know that I like Wellington and that I would do anything to have him live. I would never kill Wellington, Mrs. Shears."

"Goodbye, Christopher."

The door closed. I stood there, thinking. Mrs. Shears probably didn't kill Wellington because she was shocked that Wellington was dead. But whoever had killed him had killed him with Mrs. Shears's fork. And the shed was locked. Not only did the person who killed Wellington know that there was this shed, he or she also knew how to get it unlocked and knew that there was a fork there. So the person must have been close to Mrs. Shears.

I heard a noise and turned round and saw Mrs. Shears standing on the lawn looking at me and she threatened to call the police so I went home because I had already had a caution.

Chapter 47

I can sometimes have trouble following directions. This is because when people tell you what to do it can be confusing.

For instance, many parks have signs saying "keep off the grass". But they don't tell you what "the grass" refers to, and logically it could refer to either all the grass around that sign, all the grass in the entire park, or all the grass in the entire world. And when I asked how people know what it meant Siobhan told me that people *take their best guess*. So I try to do this, but my guesses can be wrong, and this can cause problems.

Siobhan understands. When she tells me to do something she tells me exactly what it is that I should do. I like this.

For example, she once said:

"You must never punch Sarah or hit her in any way, Christopher. Even if she hits you first. If she does hit you again, move away from her and come and tell me or one of the other members what she has done."

"If you want to go on the swings and there are already people on the swings, you must not push them off. You must ask them if you can have a go. And then you must wait until they have finished and got down the swing to use it, even if they say yes."

I asked why I have to ask, and she said that I am conveying my intent to go on the swings by asking, but I didn't know that questions can convey information instead of get information then so I was confused.

But when other people tell you what you can or can't do they don't do it like this, and I can get confused. So I *take my best guesses* even though they can be wrong, and if I feel unsure I just ask for clarification.

Chapter 53

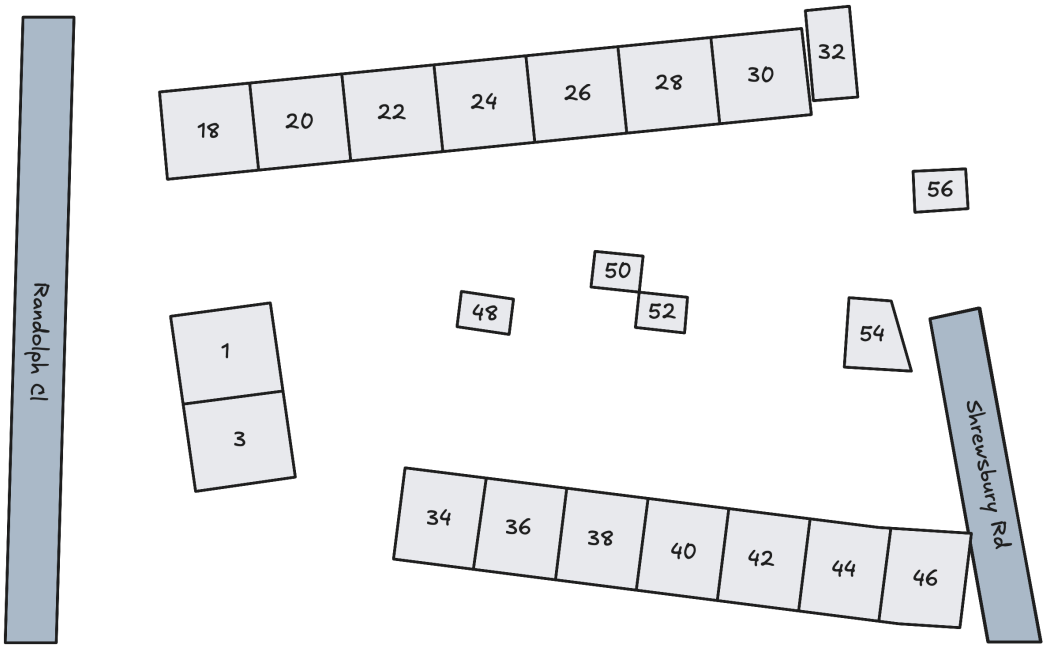
The next day was Saturday and England were playing Romania at football, which meant that we weren't going to go on an outing because Father wanted to watch the match on the television, so there is not much to do. So I decided to do some detective work.

I decided that I would go and ask some of the other people who lived in our street. Talking to strangers is not something I usually do. I do not like talking to strangers because I do not like people I have never met before since they are even more unpredictable and harder to understand.

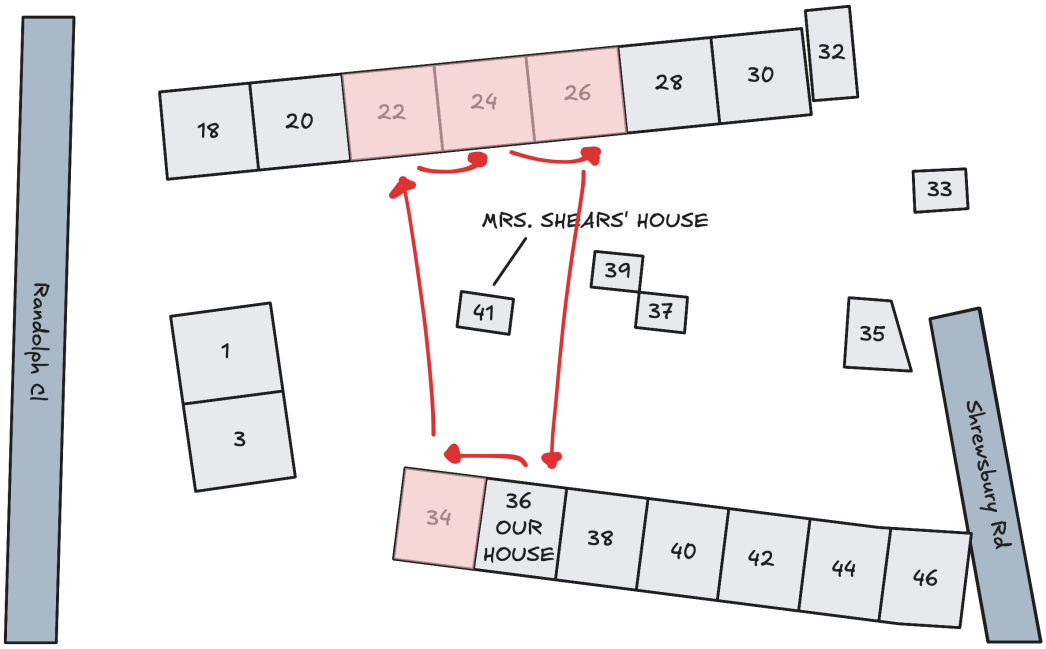
It takes me a long time to get used to people I do not know. For example, when there is a new member of staff at school I do not talk to them for weeks until I feel comfortable around them. Then I ask them questions about themselves, like whether they have pets and what is their favorite colour and what do they know about so I get to know them. Eventually, I no longer mind if I am in the same room as them and don't have to watch them all the time.

Because this is something new I am doing, I decided to *plan ahead*.

First of all I made a plan of our part of the street, like this:



Then I shaded all the houses close to Mrs. Shears' house and sketched out a path that I would follow. I didn't want to go to house 38 since Mother had told me that they take drugs and they play loud music that scares me.



So I went out and followed my plan and knocked on the door of number 34, which is opposite Mrs. Shears's house, which means that they were most likely to have seen something. The people who live at number 34 are the Thompsons.

Mr. Thompson answered the door. He was wearing a T-shirt which said

BEER

Helping ugly people have sex for 2,000 years

and I remembered that school warned us about beer and sex and that scared me so I didn't want to talk any more, but I had to find an excuse so I said that I *had the wrong number* and left.

There was no answer at house number 22.

So I went to number 24, which is the house next to Mrs. Shears's house. The people who live at number 24 are Mr. Wise and Mr. Wise's mother, who is in a wheelchair, which is why he lives with her so he can take her to the shops and take her around. It was Mr. Wise who answered the door. He smelled of body odor and old biscuits, which is what you smell of if you haven't washed for a very long time, like *Redacted* at school smells because his family is poor. (Siobhan said that I shouldn't include his name, which is why I have redact it.)

I asked Mr. Wise if he knew who had killed Wellington on Thursday night.

"Bloody hell, policemen really are getting younger, aren't they." Then he laughed.

So I thought about what he meant and guessed that he had probably thought that I was a policemen and was amused that I was so young but I didn't know how to respond so I just *laughed along*.

And then I asked if he had seen anything unusual on Thursday and he said no without thinking about it, and so I asked instead if he knew anyone who might want to have made Mrs. Shears sad.

"Perhaps you should ask your father."

"My father said he didn't know anything, and I want to figure this out by myself."

"Well, I can't think of anyone who would have wanted to make her sad."

"I'm sorry that I couldn't help you, but good luck."

I knew that that meant that I couldn't get more help from him.

And so I went to number 26, and I noticed that the old lady who lives there was in her front garden cutting her hedge with an electric hedge trimmer which made me uncomfortable since it was loud and noisy. Her name is Mrs. Alexander. She has a dog. But the dog wasn't in the garden with her, so it was probably inside the house. Whoever murdered Wellington could have also killed her dog because the murderer hated dogs but no report was made so Wellington's murder was likely more personal.

"Mrs. Alexander," I called, hoping to get her attention amidst the chaos.

She heard my speaking, so she turned the trimmer off, which made me happy.

"Do you know anything about Wellington being killed?"

"I heard about it yesterday. Dreadful. Dreadful."

This was an implicit denial.

"Do you know who killed him?"

"No, I don't."

I didn't know what else to ask, and so I thanked her for helping me with my investigation.

"It's very nice of you to come."

I slightly nodded my head to indicate acknowledgement.

"Even if it's only because you're doing detective work."

And I didn't nod back, in fear that this would *drag on* even more.

"I have a grandson your age."

I didn't know what to say so I just nodded again.

"You don't have a dog, do you?"

"No."

"You'd probably like a dog, wouldn't you."

"Yes."

"Do you want to come in for tea?"

And I saw this as a prefect opportunity to *bail out* and I told him that I didn't like tea and that I should probably be back home before Father notices that I have been out for so long.

Chapter 59

And as I was crossing the street I brainstormed about what the possible motives could be.

Why would you kill a dog?

- Because you wanted the dog gone.
  - Because the dog being dead makes your life easier.
  - Because something good will happen after the dog is dead.
  - Because it will serve as revenge.
  - Because you feel personal hatred towards the dog.
- Because something had forced you to kill the dog.
  - ...

But that didn't seem to go anywhere. And so I went home and I booted up ChatGPT and I asked it this:

> Somebody had killed my neighbor's dog with a garden fork. What are 10 reasons somebody might kill a dog?

And it responded like this:

Here are 10 possible reasons someone might kill a dog: some tragic, some malicious, and some stemming from misunderstanding. None of these justify the act, but they represent motives that might be explored in fiction or real-life investigations:

1. **Revenge or Retaliation**: The killer may have been angry at the owner and took it out on the dog.
2. **Fear or Self-Defense**: They might have believed the dog was dangerous or about to attack.
3. **Mental Illness**: The person may have been experiencing delusions, hallucinations, or psychosis.
4. **Animal Cruelty or Sadism**: Some individuals like personal gratification or to exert control or power.
5. **Accidental Escalation**: They might have intended only to scare or shoo the dog away but went too far.
6. **Jealousy or Spite**: If the dog received more attention or love, someone might be jealous or hating.
7. **Territorial Behavior**: The dog may have trespassed or damaged property, provoking violent response.
8. **Symbolic Act**: The act might have been meant as a message or threat to the owner.
9. **Noise or Disturbance**: Behavior like chronic barking could have frustrated someone into lashing out.
10. **Superstition or Misguided Belief**: In rare cases, people may act based on irrational beliefs.

If you're exploring this topic in a narrative or analytical context (like a novel or an investigation), understanding the motive is key to understanding the perpetrator.

Chapter 61

My school is a special-needs school. I get why these schools exist, but I hate the concept. So many people all belong to this "Special Needs" club, even though we don't have much in common. It is as if everybody has a place that they belong to, and people who don't is crammed into a special-needs school like a trash can for everything that doesn't belong.

But Siobhan said we have to use those words because people used to call children like the children at school nasty words. But that is stupid too because now the children from the school down the road see me on the street and they just shout "Special Needs" as if I am stupid and it is just transferring the connotation to another phrase.

Mother and Father had cared for me a lot. This was because of the stress of looking after someone who has Behavioral Problems like me. I used to have lots of Behavioral Problems, but I don't have so many now because I've rationalised them away. These are some of my past Behavioral Problems:

- Not talking to people for a long time. I've learned to forgive people for slightly infuriating me since I've learned that I can be very easily irritated and that forgiving people is not as hard as I thought.
- Not liking being touched. I still haven't come over this completely, especially when someone grabs me with force, but I like to think I am a lot better with touch.
- Screaming when I am angry or confused. I've rationalised that response away, since screaming doesn't fix things and only makes everything messier, especially for other people. Now I just overload my brain to force a reset.
- Not liking yellow things or brown things and refusing to touch yellow things or brown things. I told myself that that was irrational. But I still prefer red when there is a choice.
- Not eating food if different sorts of food are touching each other. That's gross and doesn't taste as good. I'm not changing that.
- Driving Mother's car. I've only tried to do it once when I was small and it didn't go too good. Thankfully neither me nor the car was hurt. Then Mother died and the car went away with her.
- Being stupid. I guess that *kind of sums it up.*

And so next month I'm going to take my A level in mathematics and I am going to prove that I'm not stupid and I'm just different from other people so if anyone ever comes up to me with that "Special Needs" nonsense again I'll *hit them* with my A level certificate.

Chapter 67

When I got home the next day Father was sitting at the table in the kitchen and he had made my supper. The supper was baked beans and broccoli and two slices of ham and they were laid out on the plate so that they were not touching.

"Where have you been?"

"I have been out."

"I have just had a phone call from Mrs. Shears."

That is not good.

So I started eating my baked beans and two slices of ham.

"What the hell were you doing poking round her garden?"

I continued eating my baked beans and two slices of ham.

I didn't know what to say, so I started eating the broccoli.

"How many times do I have to tell you, Christopher?"

I continued eating the broccoli.

"Keep your nose out of other people's business."

"But I liked Wellington."

Father didn't say anything.

"Honestly, I don't know if I will still be alive at this time if Wellington isn't there."

The room went silent.

And suddenly, as if having made a hard decision...

"OK, Christopher. I am going to say this for the last and final time. I will not tell you again. Look at me when I'm talking to you, for fuck's sake. Look at me. You are not to go asking Mrs. Shears about who killed that bloody dog. You are not to go asking anyone about who killed that bloody dog. You are not to go trespassing in other people's gardens. Is that understood?"

Silence.

"I am going to make you promise, Christopher. And you know what it means when I make you promise."

Silence.

I did know what it meant when you say you promise something. I don't tell lies, and that makes me exploitable.

"Promise me you will stop doing these things."

Silence.

I had already gotten all the information I had needed from inquiries. I would not need to ask anyone again. I would not need to go to Mrs. Shears' garden again. I would not benefit from not making this promise.

"I promise."

Chapter 71

The next day at school I told Siobhan that Father had told me I couldn't do any more inquiring. She said that it didn't matter. She said the book was really good as it was and that I should be very proud of having written a book at all. But I said that it wasn't a proper book because it didn't have a proper ending.

She said that was like life, and not all murders were solved and not all murderers were caught.

But my curiosity *got the better of me* and I decided that I must find the truth.

So I looked at the possibilities that were given to me by ChatGPT, and I decided that I wanted to narrow it down because some cases were either unlikely or irrelevant. So I constructed a table like so:

Reason	**Worth Considering?**
Revenge or Retaliation	**Yes.** e.g. **Mr. Shears** didn't like Mrs. Shears.
Fear or Self-Defense	**No.** I don't know anyone like this.
Mental Illness	**No.** I don't know anyone like this.
Animal Cruelty or Sadism	**No.** I don't know anyone like this.
Accidental Escalation	**No.** That wouldn't distinguish between suspects.
Jealousy or Spite	**Maybe?** Mrs. Shears did care for her dog very much.
Territorial Behavior	**No.** The dog was in Mrs. Shears' garden.
Symbolic Act	**No.** This is a subset of **Revenge or Retaliation**.
Noise or Disturbance	**No.** That wouldn't distinguish between suspects.
Superstition or Misguided Belief	**No.** I don't know anyone like this.
Summary	**Consider cases for Revenge or Jealousy.**

And then I went to bed.

Chapter 73

The next day, I was shopping and I turned round and saw Mrs. Alexander, the old lady from number 26, again. And then I saw that Mrs. Alexander's dachshund was sitting on the pavement, so I bent down and I said hello to her dog and it licked my hand which made me happy.

"His name is Ivor."

"You're very shy, aren't you, Christopher."

I nodded.

Then she tried to initiate a conversation.

"Do you like computers?"

"Yes."

Then I thought that maybe I should talk to her more, since she seemed interesting.

"I have a computer at home in my bedroom."

"I know. I can see you sitting at your computer in your bedroom sometimes."

"I also like maths and looking after Toby. And I like being on my own."

And then I had realised that what I had said was stupid because Alexander could take it the wrong way and think that I want to be on my own right now, so I hastily added:

"I'm very good at mathematics. I'm going to do my A level maths next month."

"Really? A level maths?"

I knew that since it was high-pitched, she was probably surprised. So I confirmed that that is indeed true because that is what you are supposed to do when someone is surprised.

"Well, I am very impressed. And I hope you do get an A."

"I will."

That was just being polite.

Then I thought that my conversation with Mrs. Alexander hadn't ended properly last time so maybe there was more to it, and I remembered that one of my suspects was Mr. Shears and that maybe I could ask about him. So I asked her if she knew Mr. Shears.

"Perhaps it would be best not to talk about these things, Christopher."

"Why not?"

"Because... because I think you know why your father doesn't like Mr. Shears very much."

But I didn't know. So I took my best guess and guessed that it had to do with Mother.

"Did it have to do with Mother?"

Silence.

And I knew that that meant she was hiding something.

Which meant that it *had* to do with Mother.

"Did he kill Mother?"

"Did he kill Mother?"

"No. No. Of course he didn't kill your mother."

"But did he give her stress so that she died of a heart attack?"

"What are you talking about?"

"Or did he hurt her so that she had to go into hospital?"

Silence.

"Oh, Christopher, I am so, so sorry. I never realised."

"I think you know why your father doesn't like Mr. Shears very much."

Why had she said that?

"You said that I would know why my father doesn't like Mr. Shears. Why would I know?"

Silence.

And I knew that that meant she was hiding something again.

"You said that I would know why my father doesn't like Mr. Shears. Why would I know?"

"So you don't know."

Silence.

"Christopher, look, I probably shouldn't be telling you this. Perhaps we should take a little walk in the park together. This is not the place to be talking about this kind of thing."

And I was excited so I decided to walk into the park with Mrs. Alexander.

"I am going to say something to you and you must promise not to tell your father that I told you this."

"I shouldn't have said what I said. And if I don't explain, you'll carry on wondering what I meant. And you might ask your father. And I don't want you to do that because I don't want you to upset him. So I'm going to explain why I said what I said. But before I do that you have to promise not to tell anyone I said this to you."

"Your mother, before she died, was very good friends with Mr. Shears. They were very good friends. Very, very good friends."

And so I knew that they were having sex, because adults always thought that children like me were too young for such things.

Silence.

Silence longer than anything I had heard.

Silence, as if we were moving at the speed of light and relativity had stretched the 45 seconds into what seemed to be longer than... an eternity.

"I'm sorry, Christopher. I thought you knew. That's why your father thinks that Mr. Shears is an evil man. And that will be why he doesn't want you going around talking to people about Mr. Shears. Because..."

I *cut her to the chase*. "Because he doesn't want me to know."

Silence.

"Yes."

And the pieces clicked together, like a solved Rubik's Cube finally being shown under the spotlight, all of its parts unifying to create an integral, consistent presence.

"Was that why Mr. Shears left Mrs. Shears, because he was doing sex with someone else when he was married to Mrs. Shears?"

"Yes."

"Thank you."

"If you want to talk about this you can come and see me anytime you want. You only have to knock on my door."

I nodded.

"Christopher?"

"What?"

"You won't tell your father about this conversation, will you?"

"No. I promise."

"You go on home. And remember what I said. Anytime."

Then I went home.

So Mr. Shears had left and gone to a place far away. Which meant that he was less likely to be the murderer and that I should consider other possibilities.

So I went back to the table and eliminated the **No** options and updated the table:

Reason	**Worth Considering?**
Revenge or Retaliation	**Maybe?** But I don't know of a candidate.
Jealousy or Spite	**Maybe?** Mrs. Shears did care for her dog very much.
Summary	**Consider cases for Revenge or Jealousy.**

And I thought that I should probably think of the candidates:

Reason	**Consider?**	**Candidates**
Revenge or Retaliation	Maybe?	Mr. Shears is eliminated, so I don't know.
Jealousy or Spite	Maybe?	Someone Mrs. Shears cares for less than Wellington.
Summary	—	**A friend or past friend of Mrs. Shears?**

And I went to sleep.

Chapter 79

Siobhan said that I liked mathematics because it was safe. He said I liked mathematics because it meant solving problems, and these problems were difficult and interesting but there was always a straightforward answer at the end. And that mathematics wasn't like life because in life there are no straightforward answers at the end.

But that is not true. Mathematics does not always have straightforward answers. A famous story that illustrates this is called The Monty Hall Problem.

There used to be a column called *Ask Marilyn* in a magazine called *Parade in America*, written by Marilyn vos Savant who was said to have the highest IQ in the world. And in the column she answered maths questions sent in by readers. And in September 1990 this question was sent:

> Suppose you're on a game show, and you're given the choice of three doors. Behind one door is a car, behind the others, goats. You pick a door, say #1, and the host, who knows what's behind the doors, opens another door, say #3, which has a goat. He says to you, "Do you want to pick door #2?" Is it to your advantage to switch your choice of doors?

Marilyn said that you should always change because the chances are 2 in 3 that there will be a car behind that door.

But one's intuition is that the chance is 50-50 because you don't seem to have much distinguishing information.

Lots of people wrote to say that Marilyn was wrong, even when she explained very carefully why she was right. Of the letters she got about the problem, nine out of ten readers disagreed. Here are some of the things that they said:

> Your answer to the question is in error. But if it is any consolation, many of my academic colleagues have also been stumped by this problem.

> You're in error, but Albert Einstein earned a dearer place in the hearts of people after he admitted his errors.

> I have been a faithful reader of your column, and I have not, until now, had any reason to doubt you. However, in this matter (for which I do have expertise), your answer is clearly at odds with the truth.

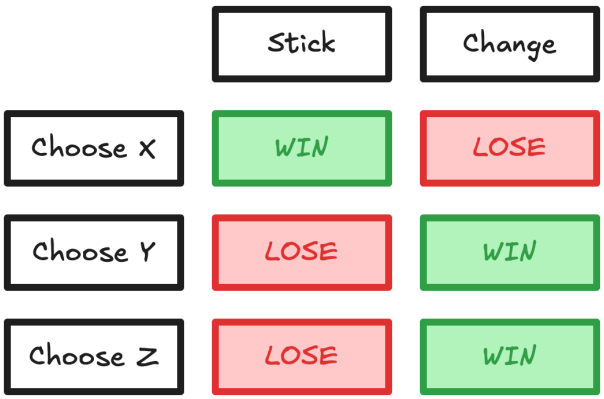
> May I suggest that you obtain and refer to a standard textbook on probability before you try to answer a question of this type again?

> You are utterly incorrect about the game show question, and I hope this controversy will call some public attention to the serious national crisis in mathematical education. If you can admit your error, you will have contributed constructively towards the solution of a deplorable situation. How many irate mathematicians are needed to get you to change your mind?
>
> Maybe women look at math problems differently than men.
>
> You made a mistake, but look at the positive side. If all those Ph.D.'s were wrong, the country would be in some very serious trouble.

But Marilyn was right, and there are many ways you can show this. For instance, you can use the Bayesian Principle, which is what I use for conditional probability like finding out how Wellington was killed:

Let the doors be called A, B, and C. Let us say that you have chosen door A. If you don't switch, $\Pr[\text{Keep gets Car}] = \Pr[\text{Car in } A] = 1/3$, which is simple. You can already conclude that you should switch, since $1/3 < 1/2$. But what most people don't understand is why $\Pr[\text{Switch gets Car}] = 2/3$. We can solve it with conditional probability: $\Pr[\text{Switch gets Car}] = \Pr[\text{Car in } Y \wedge \text{Host eliminates } Z] + \Pr[\text{Car in } Z \wedge \text{Host eliminates } Y] = 2\Pr[\text{Car in } Y] \times \Pr[\text{Host eliminates } Z | \text{Car in } Y]$ by applying the Bayesian Principle (the 2 comes from combining the symmetric terms). $\Pr[\text{Host eliminates } Z | \text{Car in } Y]$ is 1 since the Host can only eliminate Z if the car is in Y and $\Pr[\text{Car in } Y]$ is clearly $1/3$, so this works out to be $2/3$.

You can work out this probability by iterating through the entire sample space:



So if you change, 2 times out of 3 you get a car. And if you stick, you only get a car 1 time out of 3.

And this shows that intuition can be wrong, but logic can usually help you work out the right answer. It also shows that Siobhan was wrong and numbers are sometimes very complicated and not very straightforward at all.

And that is why I like The Monty Hall Problem.

Chapter 83

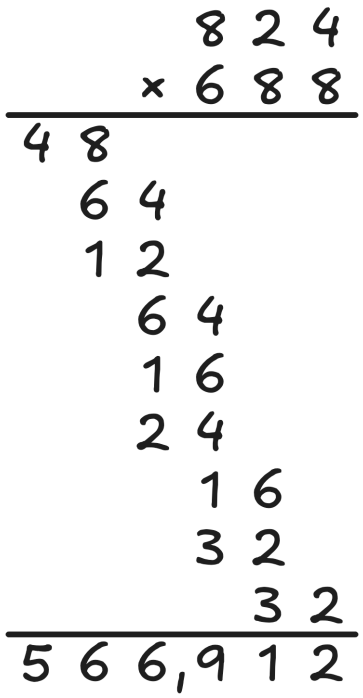
When I got home the next day Rhodri was there. Rhodri is the man who helps Father do heating maintenance and boiler repair, and he sometimes comes round to the house in the evening to drink beer with Father and watch the television and have a conversation.

Father asked what I had been up to, so I said that I went to the shop and also talked to Mrs. Alexander's dog outside the shop.

And so Rhodri asked how I was doing, and so I said that I was doing fine.

"What's 824×688?"

I mentally did what is called cross multiplication left to right so I can say the answer as I solve:



And so I said the answer and asked if that was right, and Rhodri took out his calculator and verified that it was indeed correct. And Rhodri and Father had a little conversation about the circuit boards and so I went to the garden.

Chapter 89

My memory is like a film. That is why I am good at remembering things, like the conversations I have written down in this book. When people ask me to recall something, I can simply press *Rewind* and *Fast Forward* and *Pause* like on a video recorder in my mind.

If someone asks what my mother was like, I can *Rewind* to lots of different scenes and say what she was like in those scenes. For example, I could Rewind to 2019, Independence Day when I was nine and Mother was wearing a pair of shorts made out of denim and a light blue bikini top and she was smoking cigarettes called Consulate which were mint flavor. She had finished sunbathing and had went into the water to swim and she said I should come and swim, too, but I said I didn't like swimming so she said I should just roll up my trousers and walk into the water a little way, and so I did.

"Look. It's lovely."

She swam backward and into the water and I thought a shark had eaten her and I screamed, so she stood up out of the water again and came over to where I was standing and held up her right hand and spread her fingers out in a fan.

"Come on, Christopher."

"Touch my hand."

"Listen to me, Christopher."

"You can do it."

So after a while I stopped screaming and I held up my left hand and spread my fingers out in a fan and we made our fingers and thumbs touch each other. It took several guesses for Mother to guess that I thought that a shark had eaten her.

"It's OK, Christopher. It's OK. There aren't any sharks in Cornwall."

(Except I can't remember anything before I was about 4 because I wasn't looking at things in the right way before then, so they didn't get recorded properly.)

And this is how I recognise someone if I don't know who they are. I locate a defining feature and I do a *Search* through my memories to see if I have met them before.

And this is also how I know how to act in difficult situations when I don't know what to do. If people say things which don't make sense, like, "later, alligator" or "take that" I do a *Search* and see if I have ever heard someone say this before, almost like a k-nearest-neighbors model.

And if someone is lying on the floor at school, I do a *Search* to find a picture of someone lying down with different intentions and then I compare the pictures with what is happening in front of me so I can decide whether they are just lying down and playing a game, having a sleep, or having an epileptic fit.

Other people have pictures in their heads, too. But these are what are called "imaginations", and I can do them as well, but they exist as a separate system because the pictures in my memory system are all pictures of things which really happened.

For instance, when Mother died, people say, “What would you want to say to your mother if she was here now?” or “What would your mother think about that?” which is stupid because they are treating an imagination as something that is real. I can imagine that if Mother was still here, that maybe she would have wanted me to take the A level sooner, but that doesn’t impact my decisions since that is an imagination and that I knew that Mother did die.

Chapter 97

When I got home from school the next day Father was still out at work, so I unlocked the front door and went inside and took my coat off. I went into the kitchen and put my things on the table. And one of the things was a list of all the people that might have been jealous of Mrs. Shear's dog that I had made. I can brainstorm in my head, but writing them down makes it easier. Then I made myself a raspberry milk shake and *heated it up* in the microwave and then went into my room to play Minesweeper, and I got 105 seconds, which was 7 seconds slower than my personal best (which had since improved to 98 seconds).

Father came home at 5:48:34 PM. I heard him coming through the front door. Then he came into the living room.

"Howdy, pardner."

"Hi."

It was 5:54:23 PM when Father came to my room holding the list.

"What is this?"

He sounded angry.

So I thought about how to respond.

"For fuck's sake, Christopher. For fuck's sake."

He was angry.

"What the fuck did I tell you, Christopher?"

And I remembered.

> You are not to go asking Mrs. Shears about who killed that bloody dog. You are not to go asking anyone about who killed that bloody dog. You are not to go trespassing in other people's gardens.

"Not to go asking Mrs. Shears about who killed that bloody dog. Not to go asking anyone about who killed that bloody dog. Not to go trespassing in other people's gardens. "

"Don't give me that bollocks, you little shit. What else did I say, Christopher?"

And so I remembered.

> Keep your nose out of other people's business.

"What else did I say, Christopher?"

"Come on. You're the fucking memory man."

"To keep my nose out of other people's business.

"Fuck yes. Not to go around sticking your fucking nose into other people's fucking business. And what do you do? You go around sticking your fucking nose into other people's fucking business. You go around raking up the past and sharing it with every Tom, Dick and Harry you bump into. What am I going to do with you, Christopher? What the fuck am I going to do with you?"

"I ask you to do one thing for me, Christopher. One thing."

"I didn't want to talk to Mrs. Alexander. It was Mrs. Alexander who —"

It was *worth a try.*

Then Father interrupted me and grabbed hold of my arm really hard, something I don't even have to *Search* to know that he had never done before.

So I was very surprised when he grabbed me.

I don't like it when people grab me.

I don't like being surprised, either.

So I hit him, involuntarily, like I hit the policeman when he took hold of my arms and lifted me onto my feet.

But Father didn't let go.

He was shouting.

So I hit him again.

Then, my memory ended.

It was like someone had switched me off and then switched me on again.

When I woke up, I was in the hospital, and I saw the clock which had said 3:45:15 PM.

That meant that at least 21 hours have passed.

I was lying on a hospital bed. There was a scar on my right hand and the side of my head was hurting. And Father was gone.

Three minutes later, I saw Father again.

"Don't you dare pull that shit again, Christopher. Don't you fucking dare."

And he took out his right hand and spread it out like a fan.

Chapter 101

The next day Father said he was sorry that he had hit me and he didn't mean to. Then, because it was Saturday (I cannot imagine how scared Father would be if I had been in the hospital for 45 hours instead of 21), he said he was going to take me on an expedition to show me that he was properly sorry, and that we were going to Twycross Zoo. So he made me some sandwiches because I don't like eating food from other places. And he said it would be OK because there wouldn't be too many people at the zoo because it was going to rain, which was nice since I don't like crowds and I like rain.

Then we drove to Twycross Zoo.

I had never been to Twycross Zoo before so I didn't have a route worked out in my mind before we got there, so we bought a guidebook from the information centre and I quickly skimmed through the map.

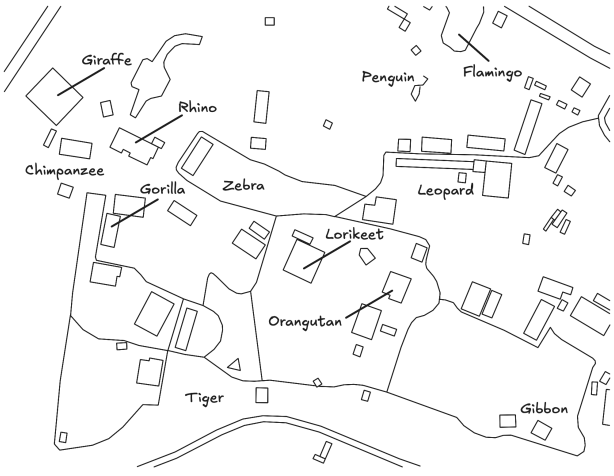
Then we went to the café where I had my sandwiches and I read the guidebook to the zoo again.

"I love you very much, Christopher. And I know I lose my rag occasionally and get angry and shout. And I know I shouldn't. I know that I make wrong decisions. And I know that I regret many of my decisions. But I only do it because I worry about you. I love you, Christopher. Do you understand?"

And I nodded because loving someone is helping them when they get into trouble, and looking after them, and telling them the truth, and Father looks after me when I get into trouble, like coming to the police station, and he looks after me, like cooking meals for me, and he always tells me the truth, which means that he loves me.

And then he held up his right hand and spread his fingers out in a fan, and I held up my left hand and spread my fingers out in a fan and we made our fingers and thumbs touch each other.

Then I got bored so I quickly sketched the map of the zoo from memory like this:



Chapter 103

I believe in God, because there is something called Pascal's Wager I found on Wikipedia and it goes as follows:

> If there is a God, He is infinitely incomprehensible, since, having neither parts nor limits, He has no affinity to us. We are then incapable of knowing either what He is or if He is.
>
> "God is, or He is not." But to which side shall we incline? Reason can decide nothing here. There is infinite chaos that separated us. A game is being played. What will you wager? You can defend neither of the propositions. The true course is not to wager at all.
>
> Yes; but you must wager. It is not optional. You are embarked. Which will you choose then? Your reason is no more shocked in choosing one rather than the other since you must of necessity choose. But your happiness? Let us weigh the gain and the loss in wagering that God is. Let us estimate these two chances. If you gain, you gain all; if you lose, you lose nothing.
>
> Wager, then, without hesitation that He is.

And Occam's Razor is a law which states

> *Entia non sunt multiplicanda praeter necessitatem.*

or literally,

> Entities must not be multiplied beyond necessity.

which translates into Bayesian language that the absurdity of the evidence must outweigh the absurdity of the claim. Only absurd evidence can qualify absurd claims.

The claim that believing in God is better may be absurd, but the absurdity of the evidence, that you gain everything and lose nothing from the belief, outweighs the absurdity of the claim, and hence this belief is grounded.

Chapter 107

When I went to school on Monday, Siobhan asked me why I had a bruise on the side of my face, so I told her about Friday night. And she asked if I was frightened about going home, and I said I wasn't. And then she asked me if I wanted to talk about it anymore, and I said that I didn't.

When the day was over, I went back home and Father wasn't there. And then I saw that the Scrabble set in Father's room had been moved, which was weird, since nobody had played. And so I thought that maybe my paper was underneath the Scrabble set, and I wanted to take it back since it was mine. So I moved away the Scrabble set, and as I moved away the Scrabble set I realised that it had been covering a cardboard box. So I opened the cardboard box and found my piece of paper.

And that was when I saw the envelope.

It was an envelope addressed to me and it was lying under my piece of paper in the shirt box with some other envelopes. And since it was addressed to me it was mine too, so I picked it up. It had never been opened. It was addressed:

Christopher Boone
36 Randolph Street
Swindon
Wiltshire

Then I noticed that underneath it were more envelopes and they were all addressed to me. And this was weird since I don't know anyone who would have written a letter to me, since I always tell them my E-mail address.

And then I heard Father opening the front door, so I took one envelope from under the book, photocopied the contents into my brain, and put it and everything else back the way they were.

"Christopher?"

I said nothing because he would be able to hear where I was calling from. I stood up and walked around the bed to the door, holding the envelope, trying to make as little noise as possible.

Father was standing at the bottom of the stairs and I thought he might see me, but he was flicking through the post which had come that morning so his head was pointing downward. Then he walked away from the foot of the stairs toward the kitchen and I closed the door of his room very quietly and went into my own room.

Chapter 109

451c Chapter Road
Wittesden
London NW2 5NG
0208 887 8907

Dear Christopher,

I'm sorry its been such a very long time since I wrote my last letter to you. I've been very busy. I've got a new job working as a secretery for a factory that makes things out of steel. You'd like it a lot. The factory is full of huge machines that make the steel and cut it and bend it into watever shapes they need. This week they're making a roof for a café in a shopping centre in Birmingham, It's shaped like a huge flower and they're going to stretch canvas over it to make it look like an enormus tent

Also we've moved into the new flat at last as you can see from the address. It's not as nice as the old one and I don't like Willesden very much, but it's easier for Roger to get to work and he's bought it (he only rented the other one), so we can get our own furnature and paint the walls the colour we want to.

And that's why it's such a long time since I wrote my last letter to you because it's been hard work packing up all our things and then unpacking them and then getting used to this new job.

I'm very tired now and I must go to sleep and I want to put this into the letterbox tomorrow morning, so I'll sign off now and write you another letter soon.

You haven't written to me yet, so I know that you are probably still angry with me. I'm sorry Christopher. But I still love you. I hope you don't stay angry with me forever. And I'd love it if you were able to write me a letter (but remember to send it to the new address!).

I also have an E-mail which is `fact.vomiting.zone@gmail.com` *if you want that.*

I think about you all the time.

Lots of Love,

Your Mum

XXXXXX

"So, what have you been up to today, young man?"

"Today we did Life Skills with Mrs. Gray. Which was Using Money and Public Transport. And I had tomato soup for lunch, and 5 apples. And I practiced some maths in the afternoon and we went for a walk in the park with Mrs. Peters and collected leaves for making collages."

"Excellent, excellent. What do you fancy for chow tonight?"

"I don't know."

Father paused.

"Then we'll have baked beans and broccoli."

Then I had my baked beans and broccoli while Father had sausages and eggs and fried bread and a mug of tea.

"I'm going to put those shelves up in the living room, if that's all right with you. I'll make a bit of a racket, I'm afraid, so if you want to watch television we're going to have to shift it upstairs."

"I'll go and be on my own in my room."

"Good man."

And I went up to my room and thought about the envelope.

I was confused because Mother had never worked as a secretary for a firm that made things out of steel. Mother had worked as a secretary for a big garage in the centre of town. And Mother had never lived in London. Mother had always lived with us. And Mother had never written a letter to me before.

So that meant the evidence was absurd, which meant that the claim will be at least equally absurd.

And then I remembered that at the front of the envelope there was a postmark and there was a date on the postmark and it was smudged and quite difficult to read, but it said



Which meant the letter was posted 18 months after Mother's supposed death.

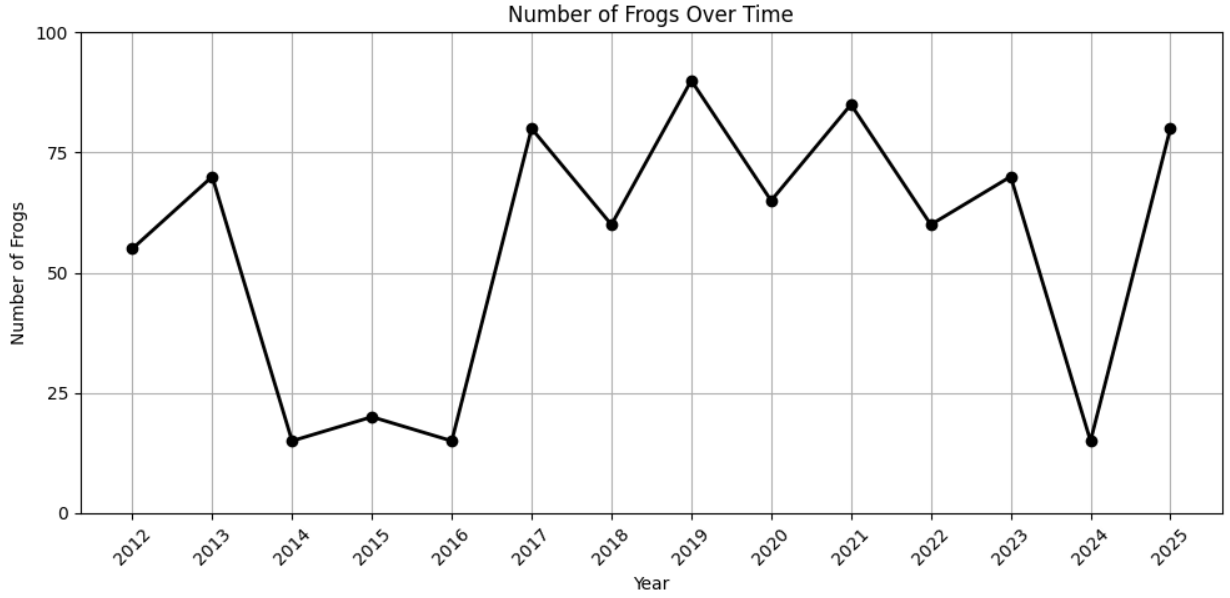
And everything clicked.

Chapter 113

Lots of things are mysteries. But that doesn't mean there isn't an answer to them. It's just that the answer hasn't been found yet.

But sometimes something that seems mysterious isn't really a mystery.

We have a pond at the school with frogs in it. And some years there are lots of frogs in the pond, and some years there are very few. And if you drew a graph of how many frogs there were in the pond, it might look like this (but this graph is only imaginary):



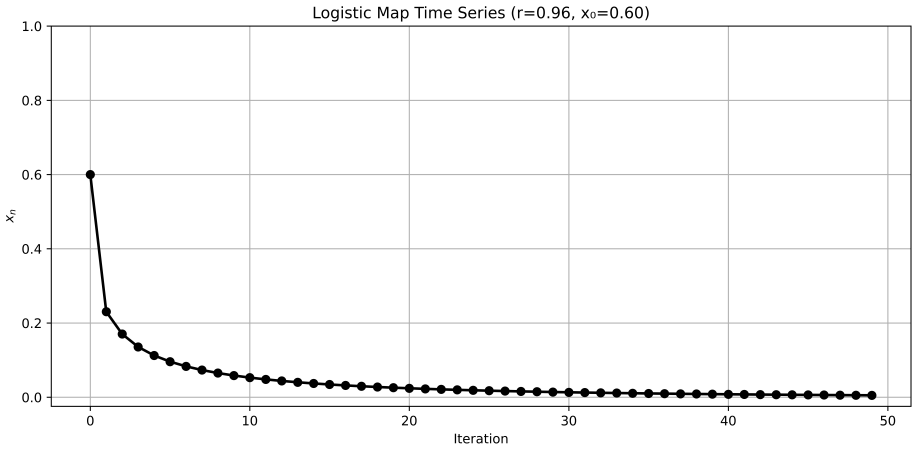
And if you looked at the graph you might think that there was a really cold winter from 2014 to 2016, or that there was a clever heron which came and ate lots of the frogs (sometimes there is a heron who comes and tries to eat the frogs, but the chicken wire over the pond usually stops it).

But sometimes there is simply no "reason". Sometimes it is just maths. Here is a simplified formula for a population of animals, not considering external factors:

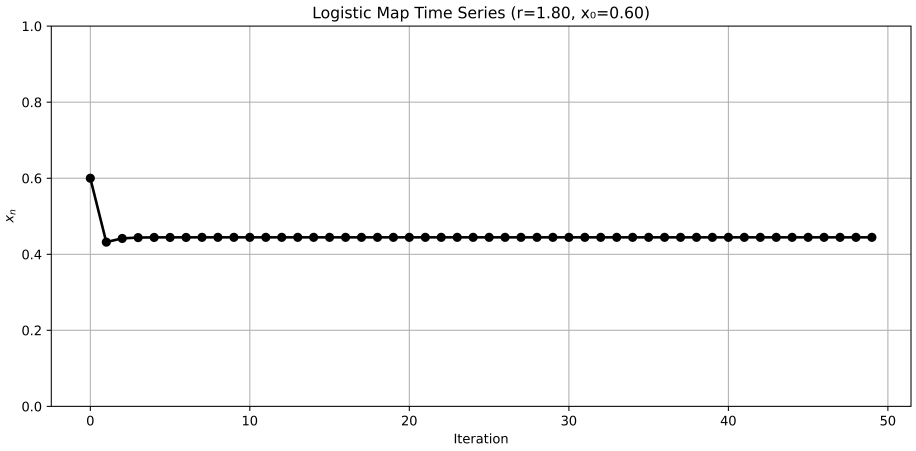
$$x' = \lambda x(1 - x)$$

Where x is the population density (not the population). When $x = 1$ the population is at carrying capacity. And when $x = 0$ the population is extirpated. x' is the population in the next year, and it is based on x, which is the current population. And λ is a parameter that influences how the population grows. To keep x' within the range of $[0, 1]$, we set λ inside the range of $[0, 4]$.

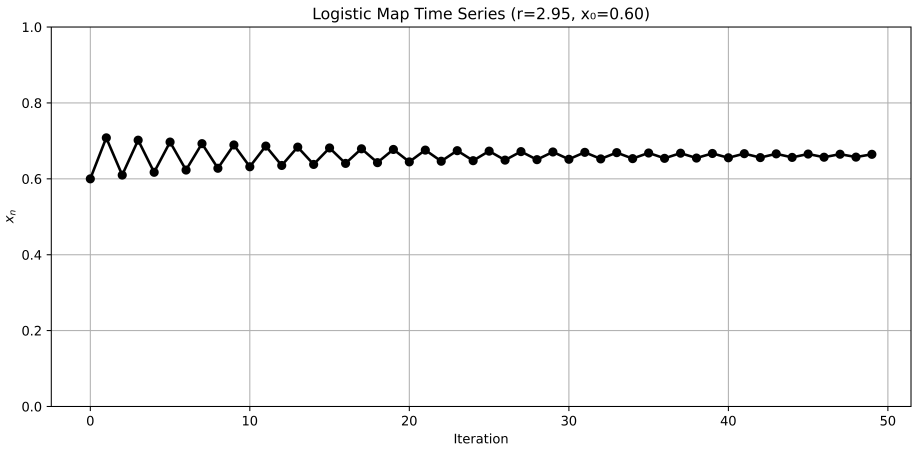
Many interesting graphs arise when we consider different values of λ.
When $0 \leq \lambda \leq 1$, the population always gradually decreases to 0.



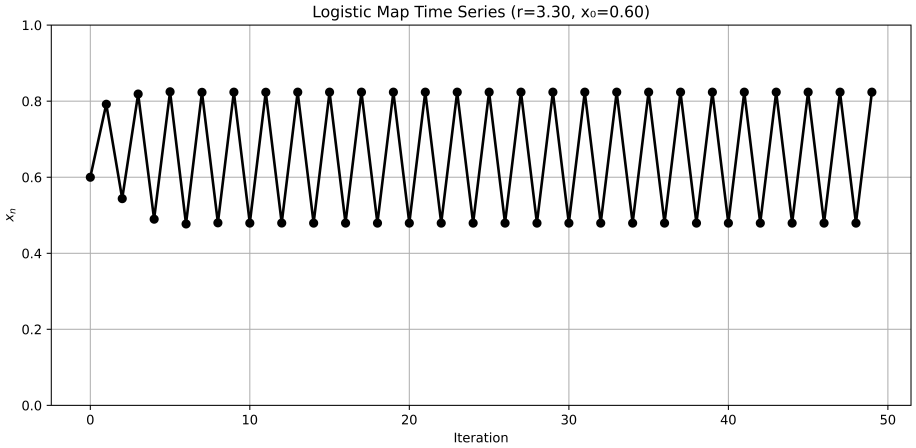
Then, when $1 \leq \lambda \leq 2$, the population quickly approaches $1 - 1/\lambda$.



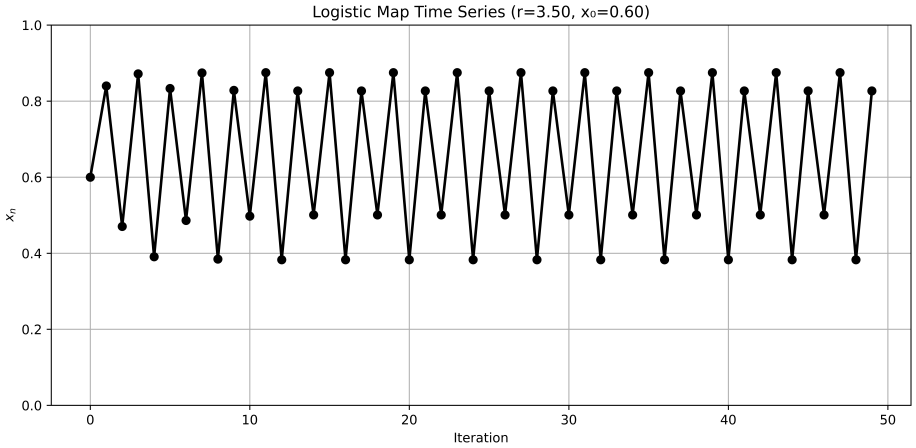
When $2 \leq \lambda \leq 3$, the population also approaches $1 - 1/\lambda$ but with an oscillation.

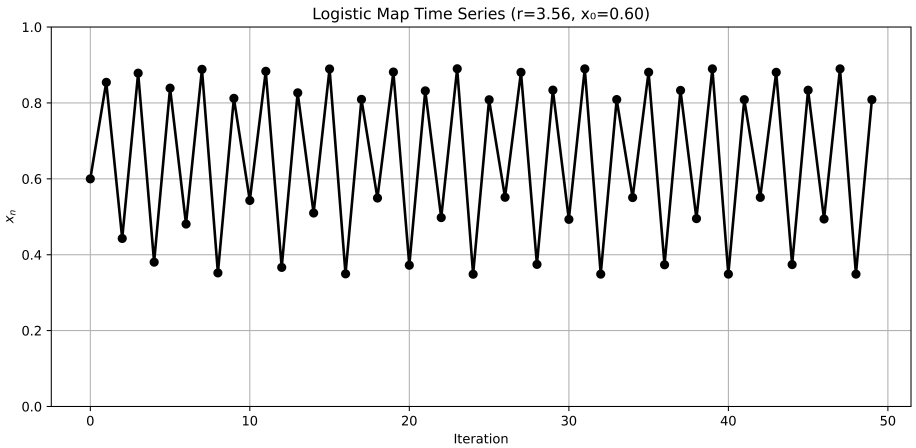


When $3 \leq \lambda \leq 1 + \sqrt{6}$, the population stabilises into an oscillation.

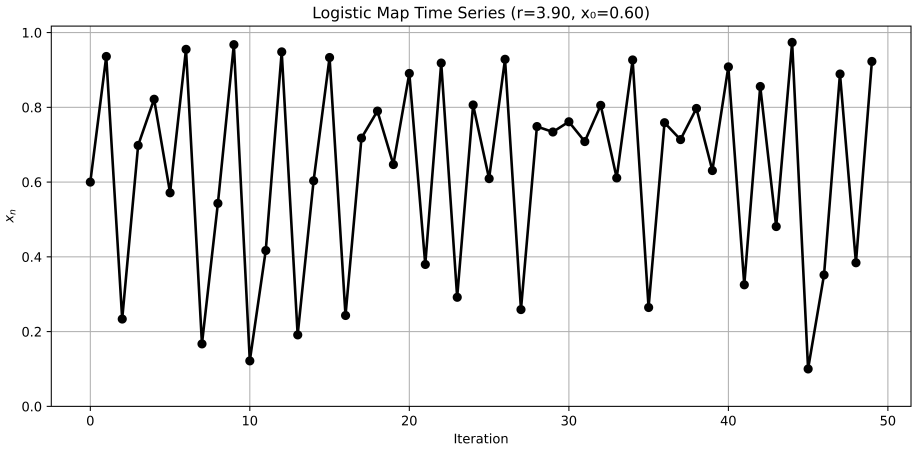


When $1 + \sqrt{6} \leq \lambda \leq \mu_\infty$, the population stabilises into a cycle with period 2^k (k is a positive integer greater than 1). μ_∞ is the critical value around 3.56995. This is called a *period-doubling bifurcation*.

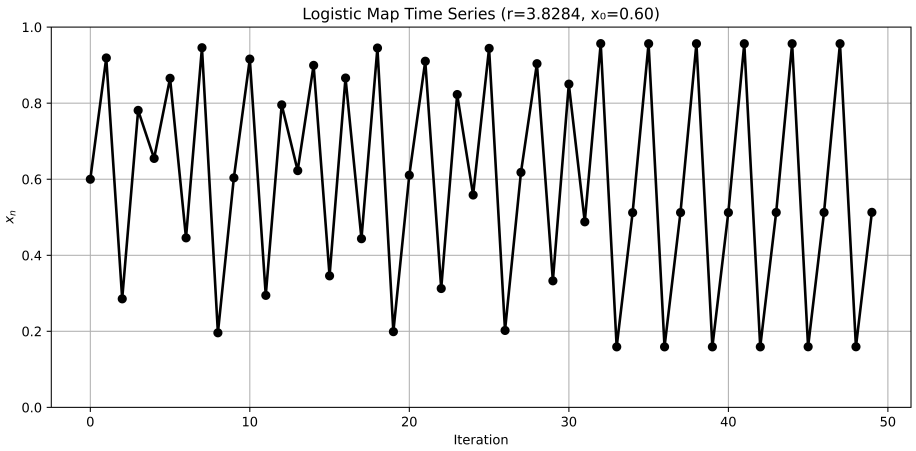




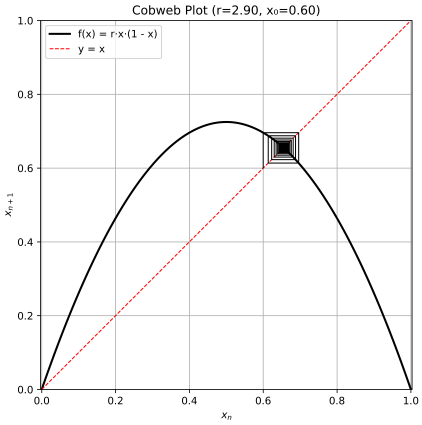
μ_∞ is called *the onset of chaos*. When $\mu_\infty < \lambda \leq 4$, the system is almost always chaotic, and so the values seem random:



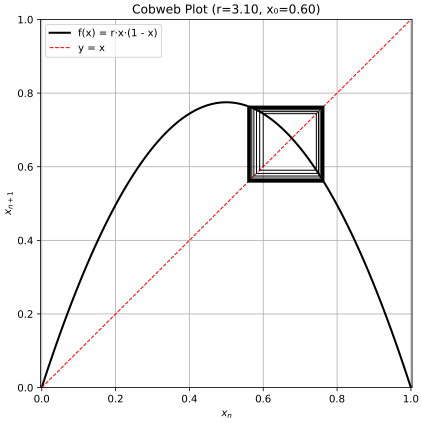
Except for some λ, like $1 + \sqrt{8}$, where there is eventually a length-3 cycle.



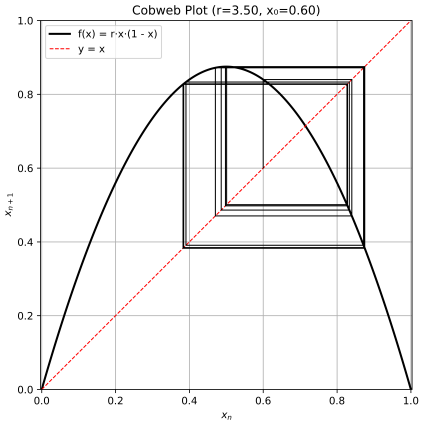
To analyze these patterns, mathematicians have created cobweb plots like these. This one converges to one point, so the series converges.



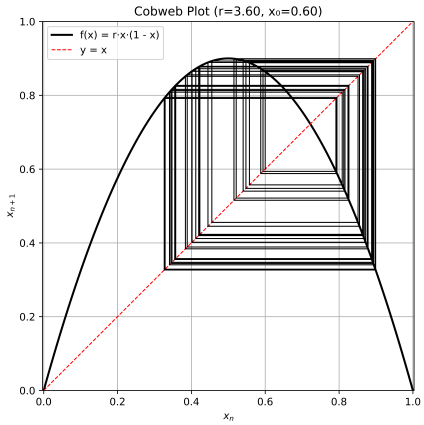
This one converges into a rectangle, so the series oscillates.



This one yields a cycle of length 4:



And chaos starts:



And so Chaos Theory states that sometimes things can seem so complicated that it is impossible to predict what they are going to do next with accuracy given imprecisions in measurement, even if they are only obeying really simple rules.

Which is to say that absurd evidence does not necessarily qualify absurd claims.

Chapter 127

I flipped back through this book.
It had all made sense.
Mother went with Mr. Shears, and Father told me she was dead.
Mrs. Shears cared for us, but not as much as her dog.
Father had killed Wellington.
It was Father.
Wellington is dead. But Mother is still alive.

Chapter 131

I took another run to Father's stash the next day, and I decided that I would look at the first letter, which would be at the bottom since his stash is in a pile or a stack, which meant that whatever came in first will come out last, and this is called *First In Last Out* or *FILO* for short.

I said that I wanted to explain to you why I went away when I had the time to do it properly. And now I do, Christopher. So I'm sitting on the sofa here with this letter and the radio on and I'm going to try and explain.

It will be hard to explain. I doubt my choices. I have made many choices, many of which are wrong, and it will do no justice to say that that particular decition was right. I do not know that it is wrong, but I do know that it is far from right.

I was not a very good mother, Christopher. Maybe if things had been different, maybe if you'd been differant, I might have been better at it. But that's just the way things turned out.

I'm not like your father. Your father is a much more pacient person. He just gets on with things and if things upset him he doesn't let it show. But that's not the way I am and there's nothing I can do to change that. I lost my temper. I don't want to lose my temper, just like how you hit people who touch you and you don't want that to to happen.

Do you remember once when we were shopping in town together? We went into Bentalls and it was really crowded and we had to get a Christmas present for Grandma, And you were frightened because of all the people in the shop. It was the middle of Christmas shopping and everybody was there and I was talking to Mr. Land who works on the kichen floor and went to school with me and you crouched down on the floor and put your hands over your ears and you were in the way of everyone and so I got cross and I told you to behave and I tried to pick you up and move you and so you shouted and you knocked those mixers off the shelf and there was a big crash and everyone turned round to see what was going on. And even though Mr. Land was really nice about it but there were boxes and bits of broken bowl on the floor and everyone was staring and I saw that you had wet yourself and I was so cross and I wanted to take you out of the shop but you wouldn't let me touch you and you just lay on the floor and screamed and banged your hands and feet on the floor and the manager came and asked what the problem was and I was at the end of my tether and I had to pay for two broken mixers and we just had to wait until you stoped screaming.

And then I had to walk you all the way home which took hours because I knew you wouldn't go on the bus again.

And I just cried and cried and cried and your father was really nice about it at first and he made you supper and he put you to bed and he said these things happen and it would be OK. But I said I couldn't take it anymore and eventually he got mad at me and he told me I was being stupid and said I should pull myself together and I hit him, which was wrong, but I was upset.

And We had a lot of argumants like that. Because I often thought I couldn't take any more. And your father is really pacient but I'm not, I get cross, even though I don't mean too. And by the end we stopped talking to each other very much because we knew it would always end up in an argumant and it would go nowere. And so I felt realy lonley. And I strated spanding time with Roger. I started seeing Roger on his own because I could talk to him. He was the only person I could really talk to. And when I was with him I didn't feel lonely anymore.

And I know you might not understand any of this, but I wanted to try to explain, so that you knew. And even if you don't understand now, you can keep this letter and read it later and maybe you might understand then.

And Roger told me that he and Eileen weren't in love with one another anymore, and that they hadn't been in love with one another for a long time. So we had a lot in common. And then we realised that we were in love with one another. And he suggested that I should leave your father and that we should move into a house together. But I said that I couldn't leave you, and he was sad about that but he understood that you were really important to me.

And then you and me had that argumant about your supper one evening. I'd cooked you something and you wouldn't eat it. And you hadn't eaten for days and days and you were looking so thin. And you started to shout and I got cross and I threw the food across the room. Which I know I shouldn't have done, but I was angry and I could not control myself. And you grabbed the chopping board and you threw that and it hit my foot and broke my toes. Then we had to go to the hospital and I had that plaster put on my foot. And afterward, at home, your father and I had a huge argumant. He blamed me for getting cross with you. And he said I should just give you what you wanted, even if it was just a plate of lettuce or a strawberry milk shake. And I said I was just trying to get you to eat something healthy. And he said you couldn't help it. And I said well I couldn't help it either.

And I couldn't walk properly for a month and your father had to look after you. And I remember looking at the two of you and seeing you together and thinking how you were really differant with him. Much calmer. And you didn't shout. And it made me so sad because it was like you didn't really need me at all. And somehow that was even worse than you and me arguing all the time because it was like I was invisible.

And I think that was when I realised you and your father were probably better off if I wasn't living in the house. Then he would only have one person to look after instead of two.

Then Roger said that he had asked the bank for a transfer. That means he asked them if he could have a job in London, and he was leaving. He asked me if I wanted to come with him.

I thought about it for a long time, Christopher.

Really, I did.

It broke my heart, but eventualy I decided it would be better for all of us if I went. So I said yes.

I meant to say goodbye, Christopher.

I was going to come back and pick up some clothes when you were back from school. And that was when I was going to explain what I was doing and say that I would come back and see you as often as I could and you could come down to London sometimes to stay with us. But when I rang your father he said I couldn't come back. He was really angry. He said I couldn't talk to you, I didn't know what to do. He said that I was being selfish and that I was never to set foot inside the house again. And so I haven't. But he said I could write letters and so I have written you these letters instead.

I wonder if you can understand any of this.

I know it will be very difficult for you.

But I hope you can understand, even if just a little bit.

I never meant to hurt you. I thought that what I was doing was the best for all of us. I hope it is. And I want you to know that this is not your fault.

I used to have dreams that everything would get better. Do you remember, you used to say that you wanted to be an astranaut? Well, I used to have dreams where you were an astranaut and you were on the television and I thought that's my son. I wonder what it is that you want to be now. Has it changed? Are you still doing maths? I hope you are.

Please, Christopher, write to me sometime, or ring me on the telephone. The numbers at the top of the letter.

Forgive me, Christopher.

And I broke into tears, right there.

And then I heard Father come up the stairs and I tried to compose myself, tried to go back to my room, tried to stop crying, tried to hide, but I couldn't do that. It was almost as if I was possessed by my crying, that my emotions had been so overwhelming that they controlled me instead of the other way around.

And so I just sat there, crying.

"Christopher, what the hell are you doing?"

And I cried some more.

"What the fuck are you... ? This is my room, Christopher. Those are... Oh shit... Shit, shit, shit, shit, shit."

Then he said nothing for a while.

Then he put his hand on my shoulder and moved me onto my side and he said, "Oh Christ." But it didn't hurt when he touched me, like it normally did. I could see him touching me, like I was watching a film of what was happening in the room, but I could hardly feel his hand at all. It was just like the wind blowing against me as I continued to cry.

"I'm sorry, Christopher. I'm so, so sorry."

"You read the letters."

And he was crying.

"I did it for your good, Christopher. Honestly I did. I never meant to lie. I just thought... I just thought it was better you didn't know... that... that... I didn't mean to... I was going to show to you when you were older."

"It was an accident."

"I didn't know what to say... I was in such a mess ... She left a note and... Then she rang and... I said she was in hospital because... because I didn't know how to explain. It was complicated. So difficult. And I... I said she was in hospital. And I know it wasn't true. But once I'd said that... I couldn't... I couldn't change it. Do you understand... Christopher...? Christopher...? It just... It got out of control and I wish..."

"Look, maybe I shouldn't say this, but... I want you to know that you can trust me. And... OK, maybe I don't tell the truth all the time. God knows, I try, Christopher, God knows I do, but... Life is difficult, you know. It's bloody hard telling the truth all the time. Sometimes it's impossible. And I want you to know that I'm trying, I really am. And perhaps this is not a very good time to say this, and I know you're not going to like it, but... You have to know that I am going to tell you the truth from now on. About everything. Because... if you don't tell the truth now, then later on... later on it hurts even more. So..."

"I killed Wellington, Christopher."

And there was silence, except for crying.

I was crying, and Father was crying beside me.

I doubled 2's in my head because it made me feel calmer and I got to 33554432, which is 2^{25}, which was not very much because I've got to 2^{236} once, but my brain wasn't working very well. And so I concentrated on 2^{236} and remembered that it was 110437941548648020598956093796432427239217753554726184882600387580688738.

"When your mum left... Eileen... Mrs. Shears... she was very good to us. Very good to me. She helped me through a very difficult time. And I'm not sure I would have made it without her. Helping out with the cooking and the cleaning. Popping over to see if we were OK, if we needed anything... I thought... Well... Shit, Christopher, I'm trying to keep this simple... but this was so complicated... I thought she might carry on coming over. I thought... and maybe I was being stupid... I was stupid... I thought she might... eventually... want to move in here. I thought we were friends. And I guess I thought wrong. I guess... in the end... it comes down to... Shit... We argued, Christopher, and... She said some things I'm not going to say to you because they're not nice, but they hurt, but... I think she cared more for that bloody dog than for us. Than for me. I mean, shit, buddy, we're not exactly low-maintenance, are we...? Anyway, we had this row. Well, quite a few rows to be honest. But after this particularly nasty little blowout, she chucked me out of the house. And you know what that bloody dog was like after the operation. Bloody schizophrenic. Nice as pie one moment, roll over, tickle its stomach. Sink its teeth into your leg the next. Anyway, we're yelling at each other and it's in the garden relieving itself. So when she slams the door

behind me the bugger's waiting for me. And... I know, I know. Maybe if I'd just given it a kick it would probably have backed off. But, shit, Christopher, when that red mist comes down... Christ, you know how it is. I mean, we're not that different, me and you. And all I could think was that she cared more about this bloody dog than she did about you or me. And it was like everything I'd been bottling up for two years just..."

And there was silence, except for crying.

Father was crying, and I was crying beside him.

"I'm sorry, Christopher. I promise you, I never meant for it to turn out like this."

"We all make mistakes, Christopher. You, me, your mum, everyone. And sometimes they're really big mistakes. We're only human."

"I'm sorry."

And there was silence, except for crying.

I was still crying.

Father was still crying.

We cried with each other.

And he took out his right hand and spread it out like a fan.

And I touched it.

Chapter 137

So I made a decision.

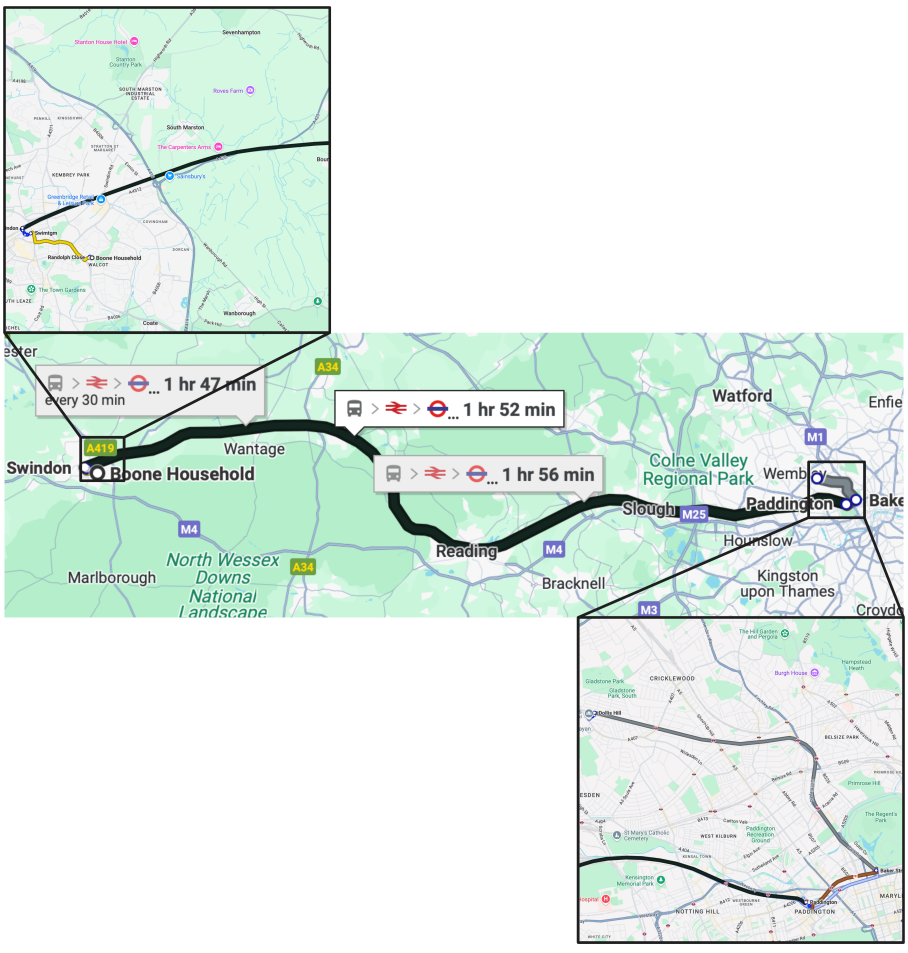
I decided that I would find out where Mother was and meet her for the first time in years.

And so I wrote an E-mail to Mother because if I didn't write such an E-mail then maybe Mother would be caught by surprise.

```
    Hi Mother,
    I know that it has been such a long time since we have met.  I want
to come visit you during the upcoming break.  I just wanted to check if
that is fine with you.

                                                              Christopher
```

Meanwhile, I went to Google Maps (wonderful website) to route my travel:



And so I noted down the steps of the plan:

- Turn right and walk along Sadler Walk.
- Turn right and enter Randolph Cl.
- Turn left into Frobisher Dr.
- Take bus 5 at Randolph Close station.
- The bus will go through Queen Dr., Fleming Way, and Corporation St.
- Leave the bus on Manchester Rd. at Orbital Shopping Pk. after 7 stops.
- Continue down and turn right into Wellington St.
- Take a GWR (Great Western Railway) to London Paddington.
- Leave the station, turn right, and go to Paddington Station.
- Take the Brown Bakerloo line through 3 stops onto Elephant & Castle on Baker Street.
- Cross Allsop Pl.
- Take the Grey Jubilee line through 7 stops onto Stanmore station on Dollis Hill.
- Leave onto Chapter Rd. and continue down.

This made me afraid, since any step could go wrong. But I decided to *run with it*.

Chapter 139

Hi Christopher,

You don't how suprised I was when I get your letter. It has been 18 months. Two years. I thought... I thought you'd never come back, never see me again. I thougth maybe the things i had done were too much, maybe those things were more unforgiveable then I thought they had been.

Yes, you can come. *Of course you can come.* Please come.

I will be neither lying nor exaggerating when I say that I had been thinking about you for the last 18 months, endlessly. Here's what I'd sent to my best friend, to help maybe reconcile the relationship between me and your father:

> its not this. its like, i fully know i want to reconcile. i fully know that if he came up to me, in any way, shape, or form, and told me to come back to his life, i will. i have imagined millions of times, countless occasions, him, walking up to me, and this family finally rebuilding. i fully know that this is what i want. yet i will not and can not allow myself to initiate, in the looming fear that whatever happens will be terribly wrong, even beyond the worst i can imagine, like an invisible black hole sucking me into the darkness. i fully know that this will not happen. i fully know that nothing can go wrong. yet my pessimism tells me that i have never done something like this, that i will never be able to do something like this. and unlike everything before, this i do not fully know to be irrational.

I'm cleaning up everything, and waiting for you to arrive. I'm happy. Tell Father that I'm ready, and I'm sorry.

Mother

Chapter 149

I see everything. I feel everything. I sense everything.

And most of the time that's good. But sometimes it can be a pain. That is why I don't like new places. If I am in a place I know, like home, or school, or the bus, or the shop, or the street, I have seen almost everything in it beforehand and all I have to do is to look at the things that have changed or moved. For example, one week the Shakespeare's Globe poster had fallen down in the classroom at school and you could tell because it had been put back slightly to the right and so three little circles of Blu-Tack stain on the wall down the left-hand side of the poster were visible. And the next day someone had graffitied *CROW APTOK* to lamppost 437 near Shrewsbury Road, the one outside number 35.

And when I am in a new place, because I see everything, it is like when a computer is doing too many things at the same time and the GPU had run out of threads and there isn't any space left to think about other things. And when I am in a new place and there are lots of people it is even harder because they can talk to you and do things that you don't expect, so you have to notice everything that is in the place, and also you have to notice things that might happen as well.

And when that happens I think of the different things that I known and am familiar with and I do several multiplication problems like 49820802×93996507 (this one was particularly easy to chunk to $49|82|08|02 \times 93|99|65|07$) at the same time so I overload my brain and force it to reboot.

And that is why I am good at board games like Scrabble, because I can memorise all the possible words and imagine every placement and every continuation and evaluate them.

Chapter 151

The next day I *hit the road* when it was 5:00:00 AM.

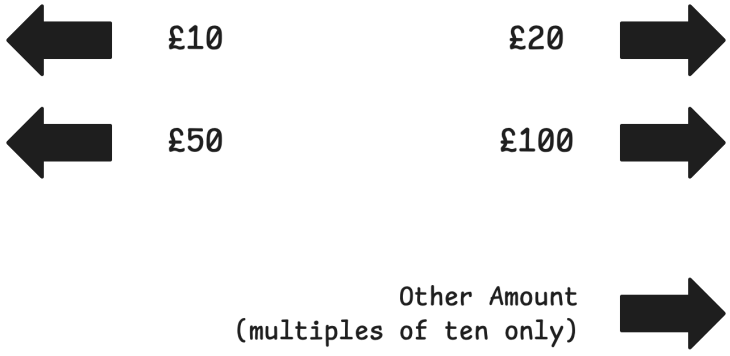
And then I walked to the station and it took me 5 minutes and I caught the bus right on time and took it to Manchester Road and paid £2 for it. At this point I had only £0.47 left, but I knew that ticket offices, especially ones in a large station, had cashpoint machines.

But I still took a look at the locations of cashpoint machines earlier and confirmed that there was, in fact, a cashpoint machine there.

And then I arrived at the train station, so I composed myself and went in.

It was like standing on a cliff in a roaring wind and it made me feel sick because there were lots of people walking into and out of the tunnel that is almost pushing me down the cliff I am about to fall down. But the ticket office was right to the left and so I went to the cashpoint machine.

I inserted the card and pressed in the pin (which I am not going to tell you) and it showed this, and I withdrew £100 since I estimated it would be £30 to £60 and I should be pessimistic since this is my first time and I don't have an inside view.



I bought a single-way ticket for 6:00:00 AM, which was the next train. It cost £32.

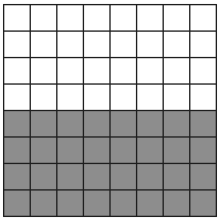
I walked into the tunnel for as long as I could and then I put my hands over my ears to block out the noise and calm myself for 30 seconds and then repeat, and eventually I got to the end of the tunnel and there were some stairs so I went up the stairs and there was a shop at the top of the stairs and a room with chairs in it but there were too many people in the room with chairs in it so I walked past it.

And there were signs saying *Great Western* and *cold beers* and *lagers* and *CAUTION WET FLOOR* and *Your 50p will keep a premature baby alive for 1.8 seconds* and *transforming travel* and *Refreshingly Different and IT'S DELICIOUS IT'S CREAMY AND IT'S ONLY £1.30 HOT CHOC DELUXE* and *0870 777 7676* and *The Lemon Tree* and *No Smoking* and *FINE TEAS* and there were some little tables with chairs next to them and no one was sitting at one of the tables and it was in a corner and so I sat down on one of the chairs next

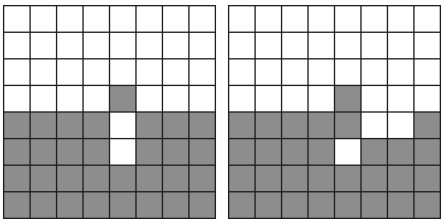
to it and I closed my eyes. And I put my hands in my pockets and Toby climbed into my hand and I gave him two pellets of rat food from my bag and I groaned to cover up the noise because I had taken my hands off my ears, but not so loud that other people would hear me groaning and come and talk to me.

And then I tried to remember what I had to do, which was to go buy a ticket, but I couldn't think then because there were too many other things in my head, so I did a maths problem to make my head clearer.

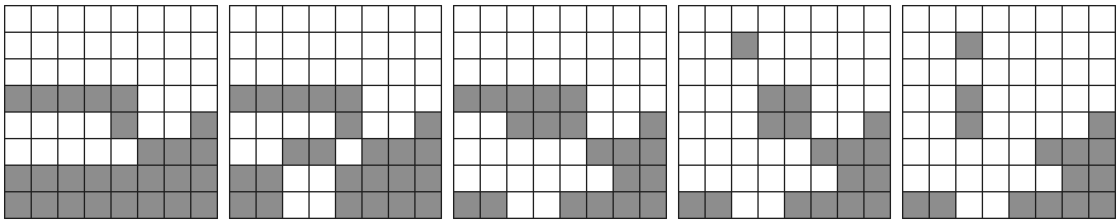
And the maths problem that I did was called Conway's Soldiers. And in Conway's Soldiers you have a finite chessboard that is very very big and filled with soldiers like this:



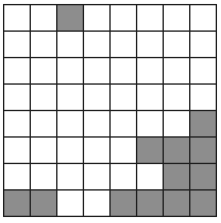
And you can move a soldier like in solitaire by jumping one over another and removing the soldier that got jumped over:



And you do that some more times like this:

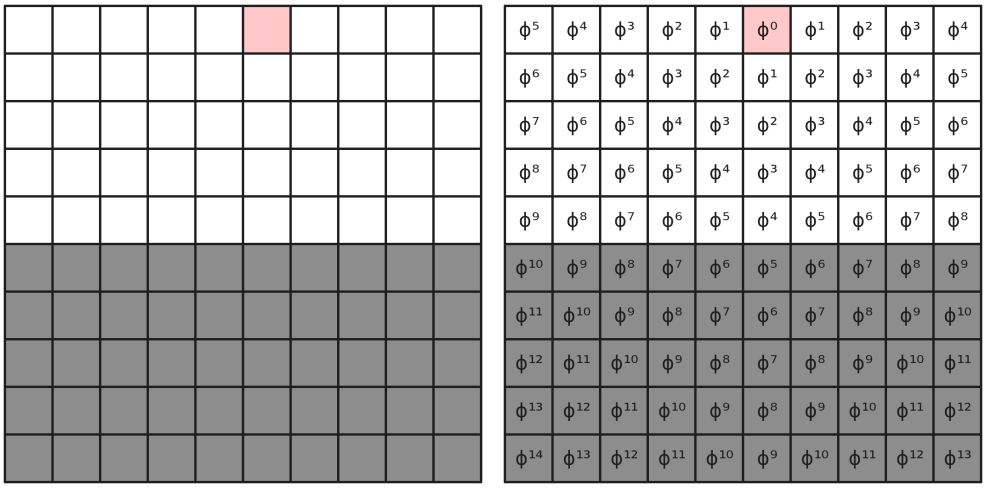


And eventually you can get a soldier to the 4th row (starting from where the highest soldiers began):



And what is amazing about this game is that you cannot get to the 5th row even if you have a billion soldiers to start with. Here's why:

We can assign each cell with a value. Suppose the target cell has weight 1. Then, each cell adjacent to it has weight φ, the golden ratio $(\sqrt{5}-1)/2 \approx 0.618$, and each cell adjacent to them weight φ^2, etc.



And we have what is called an "invariant" in mathematics, which is something that changes monotonically, and here it is the sum of the weights of all the soldiers. Due to how moving works, this invariant can only decrease. What remains to be found is the sum of the weights at the end.

Consider $S = \sum_{i=1}^{\infty} \varphi^i$. $S/\varphi - S = 1$, and $1/\varphi = 1+\varphi$, so $\varphi S = 1$. Therefore, the invariant is $2S+1$ for the target row, and hence $\varphi^4 S(2S+1)$ in total. And what comes next is simply algebraic manipulation:

$$\begin{aligned} S &= 1/\varphi \\ 2S+1 &= 2/\varphi + 1 \\ \varphi^4 S(2S+1) &= \varphi^3\,(2/\varphi + 1) \\ &= \varphi^3 + 2\varphi^2 \end{aligned}$$

Now, we use the easily verifiable identity $\varphi^2 = 1-\varphi$ to reduce the high order terms:

$$\begin{aligned} \varphi^4 S(2S+1) &= \varphi^3 + 2\varphi^2 \\ &= \varphi - \varphi^2 + 2\varphi^2 \\ &= \varphi^2 + \varphi \\ &= 1 - \varphi + \varphi \\ &= 1 \end{aligned}$$

which is just barely enough to get to our target. But since we start with a finite amount of soldiers, we hit just below 1, and so our invariant says that we will stay below 1, and so we'll never get a soldier on top of 1.

Chapter 157

And then I looked up and saw that there was a policeman in front of me.

"Anyone at home?"

I didn't know what that meant.

"Are you all right, young man?"

And I knew the standard response.

"Yes."

"You're looking a bit worse for wear."

"Yes."

"The lady at the café says you've been here for five minutes, and when she tried talking to you, you were in a complete trance."

"Yes."

"What's your name?"

"Christopher."

"Where do you live?"

"Willesden."

And I wondered whether I should tell him that Father killed Wellington and whether he would arrest Father, but then I knew that there would be better times.

"What are you doing here?"

"I needed to sit down and be quiet and think."

"OK, let's keep it simple. What are you doing at the railway station?"

"I'm trying to go to London."

"Why?"

"I want to visit Mother."

"So you don't live with your mother?"

"No. But I'm going to."

"So where does your mother live?"

"In London."

"Yes, but where in London?"

"Chapter Road."

"Jesus. What is that?"

"That's my pet rat, Toby,"

"A pet rat?"

"Yes."

"Have you got a ticket?"

"Yes."

"Now, are you sure you know what you're doing?"

"Yes."

"Has your mother got a telephone number?"

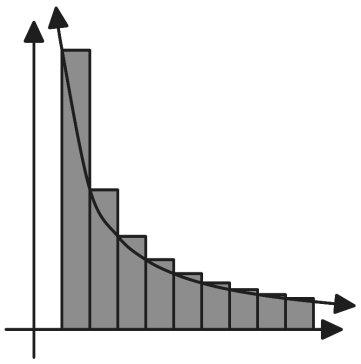
"Yes."
"And can you tell me what it is?"
"Yes."
"What is it?"
"It's 0208 887 8907."
"And you'll ring her if you get into any trouble, OK?"
"Yes."
"Good."
And there was some time so I thought about the Riemann ζ function.
Define

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$$

for real $s > 1$.

Theorem 1. $\sum_{n=1}^{\infty} n^{-1}$ *diverges.*

Proof. Use the integral test. Specifically, the area covered by the integral $\int_1^{\infty} 1/x \, \mathrm{d}x$ is strictly smaller than $\sum_{n=1}^{\infty} n^{-1}$, but $\int 1/x \, \mathrm{d}x = \ln x + C$ and hence $\int_1^{\infty} 1/x \, \mathrm{d}x = \ln x \Big|_{x=1}^{x=\infty} = \ln \infty - \ln 1 = \infty$, so $\sum_{n=1}^{\infty} n^{-1} \geq \infty$ if we *abuse notation.*



□

Theorem 2. ζ *is well-defined.*

Proof. That is, we want to prove that $\zeta(s)$ converges for all real $s > 1$. We use the integral test, since $\int x^{-s} \, \mathrm{d}x = x^{1-s}/(1-s)$ for real $s > 1$ so $\int_1^{\infty} x^{-s} \, \mathrm{d}x$ exists, and hence $\zeta(s)$ exists. □

Theorem 3. $\zeta(2) = \pi^2/6$.

Proof. Consider $f(x) = \sin x$. The Weierstrass factorisation theorem states that we can "factor" this into

$$ax \prod_{k=1}^{\infty} (1 + x/k\pi)(1 - x/k\pi)$$

and this will coincide with the Taylor expansion of $\sin x$ which is

$$\sum_{n=0}^{\infty} \frac{\sin n\pi/2 \cdot x^n}{n!} = x - x^3/6 + x^5/120 - \cdots$$

Therefore, by looking at the linear term, we find that $a = 1$, so the factorisation is

$$x \prod_{k=1}^{\infty} (1 + x/k\pi)(1 - x/k\pi) = x \prod_{k=1}^{\infty} (1 - x^2/k^2\pi^2)$$

and by looking at the cubic term, we get that

$$\sum_{k=1}^{\infty} -1/k^2\pi^2 = -1/6 \iff \zeta(2) = \sum_{k=1}^{\infty} k^{-2} = \pi^2/6.$$

□

Theorem 4. $\zeta(4) = \pi^4/90$.

Proof. Recall that

$$x \prod_{k=1}^{\infty} (1 - x^2/k^2\pi^2) = \sum_{n=0}^{\infty} \frac{\sin n\pi/2 \cdot x^n}{n!} = x - x^3/6 + x^5/120 - \cdots$$

By looking at the quintic term, we get

$$\sum_{i=1}^{\infty} \sum_{j=i+1}^{\infty} i^{-2} j^{-2} = \pi^4/120.$$

Then, consider

$$\zeta(2)^2 = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} i^{-2} j^{-2} = \pi^4/120 + \pi^4/120 + \sum_{i=1}^{\infty} i^{-4} = \pi^4/60 + \zeta(4).$$

Hence,

$$\zeta(4) = \zeta(2)^2 - \pi^4/60 = \pi^4/36 - \pi^4/60 = \pi^4/90.$$

□

Theorem 5. $\zeta(s) = \prod_{prime\ p} 1/(1-p^{-s})$.

Proof. We use the Fundamental Theorem of Arithmetic to get

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \sum_{a_i=0}^{\infty} \left(\prod_{i=1}^{\infty} p_i^{a_i} \right)^{-s} = \sum_{a_i=0}^{\infty} \prod_{i=1}^{\infty} (p_i^{a_i})^{-s} = \prod_{i=1}^{\infty} \sum_{a_i=0}^{\infty} (p_i^{a_i})^{-s} = \prod_{\text{prime } p}^{\infty} \sum_{a=0}^{\infty} \left(p^{-s}\right)^a$$

where p_i denotes the i-th prime. Therefore, what remains to be proved is that

$$\sum_{a=0}^{\infty} \left(p^{-s}\right)^a = 1/(1-p^{-s})$$

which is trivial. □

Theorem 6. *There are infinitely many primes.*

Proof. Consider

$$\zeta(2) = \pi^2/6.$$

Assume, for contradiction, that there is only a finite number of primes. Then

$$\zeta(2) = \prod_{\text{prime } p} 1/(1-p^{-2})$$

is a product of a finite amount of rational numbers. However, since π is transcendental, $\pi^2/6$ is irrational. So a rational number is irrational, which is absurd. Hence there are infinitely many primes. □

Theorem 7. *The probability that two random positive integers are co-prime is* $6/\pi^2$.

Proof. The probability that two such numbers share a common prime factor of p is p^{-2}, so the probability that they do not share the common prime factor of p is $1-p^{-2}$. Since this is independent for each prime factor, the probability that the two numbers are co-prime is

$$\prod_{\text{prime } p} (1-p^{-2}) = 1/\zeta(2) = 6/\pi^2.$$

□

Now we extend the definition of ζ to all complex s by considering the analytical continuation of the above definition. We get the functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin(s\pi/2) \Gamma(1-s) \zeta(1-s).$$

Notably, Γ is an extension of the factorial function. For non-negative integer n, $n! = \Gamma(n+1)$.

Theorem 8. $\zeta(s) = 0$ *for negative even integer* s*.*

Proof. We directly use

$$\zeta(s) = 2^s \pi^{s-1} \sin(s\pi/2)\Gamma(1-s)\zeta(1-s).$$

Notice that $\sin(s\pi/2) = 0$, so $\zeta(s) = 0$. Negativity is necessary, since $\Gamma(1-s)$ is not well-defined for positive s and $\zeta(1)$ does not exist for $s = 0$, which fails the argument.

These roots of ζ are called the trivial roots. □

Theorem 9. $\Gamma(x)\Gamma(1-x) = \pi \csc \pi x$*.*

Proof. Recall that

$$\zeta(s) = 2^s \pi^{s-1} \sin(s\pi/2)\Gamma(1-s)\zeta(1-s).$$

Similarly,

$$\zeta(1-s) = 2^{1-s} \pi^{-s} \sin((1-s)\pi/2)\Gamma(s)\zeta(s).$$

Multiplying, we get

$$\cancel{\zeta(s)\zeta(1-s)} = 2^1 \pi^{-1} \sin(s\pi/2)\sin(\pi/2 - s\pi/2)\Gamma(s)\Gamma(1-s)\cancel{\zeta(s)\zeta(1-s)},$$

so

$$\Gamma(s)\Gamma(1-s) = \pi/(2\sin(s\pi/2)\cos(s\pi/2)) = \pi/\sin(\pi s) = \pi \csc \pi x.$$

□

Theorem 10. $1 + 2 + 3 + \cdots = -1/12$*.*

Proof. The left side $1 + 2 + 3 + \cdots$ is $\zeta(-1)$. We again use

$$\zeta(s) = 2^s \pi^{s-1} \sin(s\pi/2)\Gamma(1-s)\zeta(1-s).$$

Plugging in $s = -1$, we get

$$\zeta(-1) = 2^{-1} \pi^{-2} \sin(-\pi/2)\Gamma(2)\zeta(2).$$

$\sin(-\pi/2) = -1$. $\Gamma(2) = 1! = 1$. $\zeta(2) = \pi^2/6$. Hence, $\zeta(-1) = -1/12$.

But $1 + 2 + 3 + \cdots$ doesn't actually converge, and definitely does not equal $-1/12$. The trick is that the definition for $\zeta(s)$ is no longer $\sum_{n=1}^{\infty} n^{-s}$ when $s < 1$, and so we are only analyzing the analytical continuation of ζ. □

Theorem 11. *The zeroes* s *of* ζ *have either* $Im(s) = 0$ *or* $Re(s) = 1/2$*.*

Proof. Left as an exercise for the reader. □

Chapter 163

When I was done I looked at the time again and I saw that I had lost track of time, so I rushed to platform 1. So I pictured in my head a big red line across the floor which started at my feet and went through the tunnel and I started walking along the red line, saying, "left, right, left, right, left, right," because it helps if I do something that has a rhythm to it.

Then I saw a sign saying ← *Platform 1* and the ← was pointing at a glass door so I went through it, and someone bumped into me again with a suitcase and so I made a barking noise.

And I saw a man with a newspaper and a bag of golf clubs go up to one of the doors of the train and press a big button next to it and the doors were electronic and they slid open and I liked that. And then the doors closed behind him.

And then I went up to the door and I pressed the big button and the doors slid open and I stepped through the doors, and that meant I was on the train to London.

There were lots of people on the train, and I didn't like that, because I don't like lots of people I don't know and I hate it even more if I am stuck in a room with lots of people I don't know. And a train is exactly that.

So I stood very still in the train carriage and didn't move.

"Christopher."

"Caught you just in time."

"We've got your father at the police station."

"He's looking for you."

"Yes."

"So, why are you going to London?"

"I'm going to live with Mother."

"Well, I think your father might have something to say about that."

But I couldn't get caught by the police and get taken back to Father.

So I started to run away.

He grabbed me.

I screamed.

He let go.

"OK, let's not get overexcited here."

"I'm going to take you back to the police station and you and me and your dad can sit down and have a little chat about who's going where."

But I have already decided to go.

"I'm going to London."

"Not just yet you're not."

And I saw that the train was set to go.

"Have you arrested Father?"

'Arrested him? What for?"

“He killed...”

I stumbled.

Intentionally.

“He... he killed... he killed a dog. With... with a garden... a garden fork. The... the dog was called Well... Wellington. There was... was bl... blood and everything.”

“Did he now?”

“Yes, he did. He... he killed... killed a dog. Did you arrest... arrest him?”

“Well, we can talk about that as well. Let’s go.”

He reached out.

I screamed.

“Now listen, you little monkey. You can either —”

“Shitting fuck.”

The train started to move.

“Rob... ? Yeah, it’s Nigel. I’m stuck on the bloody train. Yeah. Don’t even... Look. It stops at Didcot Parkway. So, if you can get someone to meet me with a car... Cheers. Tell his old man we’ve got him but it’s going to take a while, OK? Great.”

“Let’s get ourselves a seat.”

“Park yourself. And no fooling around.”

And the people who were sitting on the seats got up and walked away.

“You are a bloody handful, you are. Jeez.”

And outside the window was like a map, except that it was in three dimensions and it was life-size because it was the thing it was a map of. And there were so many things it made my head hurt, so I closed my eyes, but then I opened them again because it was like flying, and then the countryside started and there were fields and cows and horses and a bridge and a farm and that made me think that there must be millions of miles of train track in the world and they all go past houses and roads and rivers and fields, and that made me think how many people must be in the world and this made my head hurt again so I closed my eyes again and did some maths practice in my head.

So I thought about how 2025, which is the current year, has the property that $(20+25)^2 = 2025$. So I wondered if there were other such four-digit numbers and there were, so I tried all the squares from 1024 to 9801 and found that 2025, 3025, and 9801 were the only squares that worked. And so I wondered what eight-digit numbers have this property.

Formally, solve the Diophantine Equation $10000a + b = (a + b)^2$ with the restrictions that $1000 \leq a \leq 9999$ and $0 \leq b \leq 9999$.

We analyze this equation under moduli 9, 11, and 101. $a+b \equiv (a+b)^2$, so $(a+b)(a+b-1) \equiv 0$, and so $a + b \equiv 0$ or $a + b \equiv 1$.

The possible values of $a + b$ is therefore 0 or 1 modulo 9, 11. If we consider modulo 99, this is 0, 1, 45, or 55. Finally, if we consider modulo 9999, this is 0, 5050, 7272, 7777, 4950, 1, 2223, and 2728. We know that $(a + b)(a + b - 1)/9999 = a$, so $\sqrt{9999000} \leq a + b \leq 9999$. Therefore, the values for $a + b$ are 4950, 5050, 7272, 7777, and 9999.

Then I wanted to go wee, so I left. (I knew that trains had toilets because I read of it.)

“Christ, where are you going?”

“To the toilet.”

“For God’s sake, go to the bloody toilet.”

So I went to the toilet and did my business.

Then I went out of the toilet and I saw that opposite the toilet there were two shelves with cases and a rucksack on them and it made me think of the airing cupboard at home and how I climb in there sometimes and it makes me feel safe. So I climbed onto the middle shelf and I pulled one of the cases across like a door so that I was shut in, and it was dark and there was no one in there with me and I couldn't hear people talking so I felt much calmer and could finally think freely again and it was nice.

And I finished up the problem. Since $a = (a+b)(a+b-1)/9999$, I can just solve with my values of $a+b$, which were 4950, 5050, 7272, 7777, and 9999. And so I get

$$\begin{cases} a+b=4950 \\ a=2450 \\ b=2500 \end{cases}, \begin{cases} a+b=5050 \\ a=2550 \\ b=2500 \end{cases}, \begin{cases} a+b=7272 \\ a=5288 \\ b=1984 \end{cases}, \begin{cases} a+b=7777 \\ a=6048 \\ b=1729 \end{cases}, \begin{cases} a+b=9999 \\ a=9998 \\ b=0001 \end{cases}$$

so there are exactly five solutions, and the only such eight-digit numbers are 24502500, 25502500, 52881984, 60481729, and 99980001.

And then the train started to slow down and someone came and stood near the shelf and knocked on the door of the toilet.

"Christopher...? Christopher...?"

"Bloody hell."

He was really close so that I could see his walkie-talkie.

But he didn't see me.

So I didn't say anything.

And then he went away again, running.

Then the train stopped.

I knew London was still far off.

"You scared the living daylights out of me."

"I think someone's out there on the platform looking for you."

"Well, it's your lookout."

She went away.

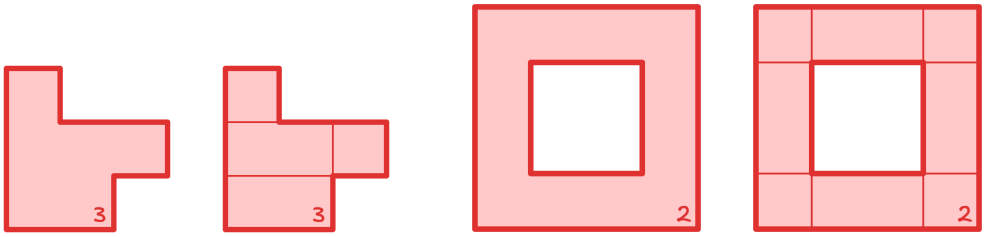
Then three other people walked past.

And then the train started going again.

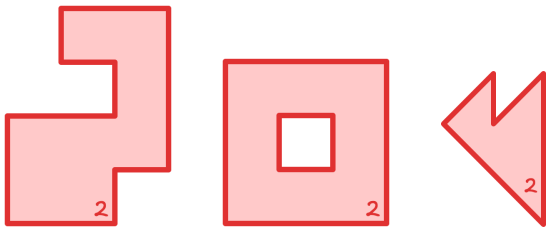
Chapter 167

And then I closed my eyes and did some more maths puzzles so I didn't think about where I was going so I did a set of origami puzzles that an internet friend had referred me to.

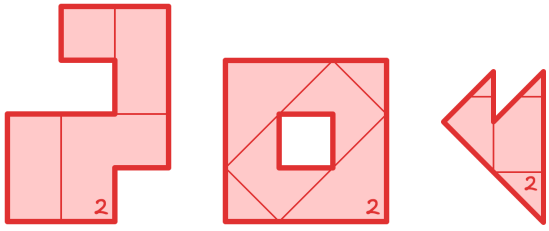
The rules were simple. Each shape has a designated thickness. I want to fold each shape into a rectangle with the given thickness. The first set of puzzles was easy because it was supposed to be introductory:



The train ran.
The train stopped.
Twyford, according to the time.
The train ran.
Then I looked at the next set, which was more interesting.



This set tells you there can be creases within the rectangle, that there can be diagonal creases, and that there can be creases that doesn't pass through lattice points. They can be solved like this:



Then there were another 4 stops and 4 people came and took bags away from the shelves and 2 people put bags on the shelves, but no one moved the big suitcase that was in front of me.

"You are fucking weird, mate."

"Have you touched this?"

"Yes."

And then she went away.

"Come and look at this, Barry. They've got, like, a train elf."

"Well, we have both been drinking."

"Perhaps we should feed him some nuts."

"You're the one who's bloody nuts."

I laughed.

They went away.

Then the train was really quiet and it didn't move again. It was London.

So I decided to get off the shelf and go and get my bag and see if the policeman was still sitting in his seat. So I got off the shelf and I looked through the door, but the policeman wasn't there. So I got off the train.

And when I saw how big the room was that the train was in and I heard how noisy and echoey it was, I had to kneel down on the ground for a bit because I thought I was going to fall over. And when I was kneeling on the ground I worked out which way to walk. So I stood up and I imagined that there was a big red line on the ground which ran parallel to the train to the gate at the far end and I walked along it with my rhythm like before.

"I think someone's looking for you, mate."

"A policeman."

"Yes."

"Oh. Right."

"You wait here, then, and I'll go and tell them."

I carried on walking, and I could still feel the feeling like a balloon inside my chest and it hurt and I covered my ears with my hands and I went and stood against the wall of a little shop which said *Hotel* and *Theatre Reservations Tel: 0207 777 5164* in the middle of the big room and then I took my hands away from my ears and I groaned to block out the noise and I looked round the big room at all the signs to see if this was London. And the signs said

Sweet Pastries Heathrow Airport Check-in Here Bagel Factory EAT excellence and taste YD! sushi Stationlink Buses W H Smith MEZZANINE Heathrow Express Clinique First Class Lounge FULLERS easyCar.com The Mad Bishoy and Bear Public House Fuller's London Pride Dixons Our Price Paddington Bear at Paddington Station Tickets Taxis + t Toilets First Aid Eastbourne Terrace lington Way Out Praed Street The Lawn Q Here Please Upper Crust Sainsbury's Local Information GREAT WESTERN FIRST Position Closed Position Closed Position Closed Sock Shop Fast Ticket Point Millie's Cookies Coffee AIRLINERS COLLIDE OVER INDONESIA: 350 FEARED DEAD Freshly Baked Cookies and Muffins Cold Drinks Penalty Fares Warning Savoury Pastries Platforms 9-14 Burger King Fresh Filled! the reef café bar business travel special edition TOP 75 ALBUMS Evening

But after a few seconds they looked like this

bliH cousFe uelrl'soLd nonPerdi iDnxos u OrrPicaedP or mnsR EAT GWUSEMSE TvEN RIFTSR PotisilCoosedS okcS o phasF ticTekt Pinot MiillseC' ookie soCfef eAIRLNEIS RODLLCIEOVR E INONDSEA:I3 F0E5 AER DDEAeDF rysh lBrow Erpxes sCnliquieF isr Ctals sonLuge LUFLRSE aeysaCr.oc Tmhe M aBsdihoa nyd Ber naSwN N IEetyH hat OruPaedtS er eTthLwea neQ deH t!hre eerei Hseratoh wAirrotpe kCih-nc eerBH ae glFctao ryETAce xeidllngotne Barat aPd doingt nSttiaoT enir PlaeesUpa tylaF resaW nprre rCustSa isunbr'ys clLanoo Imfrtai-Pu9o1 Bu4ger rKnig eF rshFiclkl etas Tx+ist TeolitFsi r stAdEi bstaounTre erracel ntiognak aWe dCooieaskn du Mffins olCD drnkiPsn eiSgn oavuyr aPsteris tPlaffcfaé b r absuniests ete Paentseca dnt ats! YeDu sihts itaSonnl kuisBe WsH iS t EmhMZA ZonlCsePd soniot Ciloeds oPisitroeva slpecia lediton TiPO 7A 5eBLinn g

because there were too many signs and my brain wasn't working properly and this frightened me so I closed my eyes again and I counted slowly to 50 but without doing the cubes. And I stood there and I opened my Swiss Army knife in my pocket to make me feel safe and I held on to it tight.

And then I made my hand into a little tube with my fingers and I opened my eyes and I looked through the tube so that I was only looking at one sign at a time and after a long time I saw a sign that said Information and it was above a window on a little shop.

"You look lost."

I instinctively took out my Swiss Army knife.

"Whoa. Whoa. Whoa. Whoa. Whoa."

And then he held up both his hands to resign and backed off.

I went to the shop that said *Information*.

"Is this London?"

"Is this London?"

"Sure is, honey."

And so I left.

Chapter 173

I took a right and entered Paddingson Station.

I saw that people were putting tickets into the gates and walking through, and some of them were buying tickets at big black machines on the wall.

So I put in £2.20.

Then, I pressed *adult single*.

Then, I put in the £2 I got from change earlier and the 20 cent change I had.

Then, I took the ticket.

Then, I went to the gate.

Then, I put my ticket in.

Then, it came out the other end.

Then, I took it.

"Get a move on."

So I walked forward and the gate opened this time.

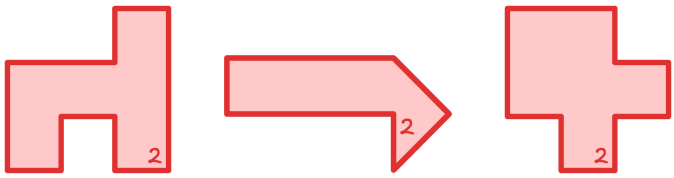
I went through the left-hand tunnel and there was a fence down the middle of the tunnel. The people were walking straight ahead on the left and coming the other way on the right like on a road, so I walked along the left and the tunnel curved left and then there were more gates and a sign said Bakerloo Line and it pointed down an escalator, so I had to go down the escalators and I had to hold on to the rubber rail which moved so I didn't fall over.

People were standing close to me and I wanted to hit them to make them go away but I didn't because they would be sad, and then I was at the bottom of the escalators and I had to jump off and I tripped and bumped into someone.

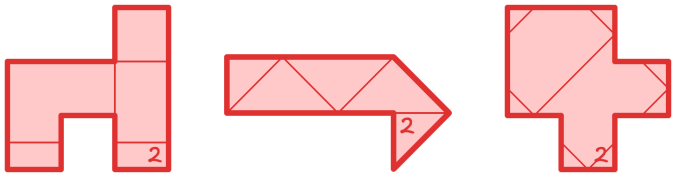
"Easy."

I was in another train station but it was tiny and it was in a tunnel and there was only one track and the walls were curved and they were covered in big adverts and they said *WAY OUT* and *London's Transport Museum* and *Take time out to regret your career choice* and *JAMAICA* and *British Rail* and *No Smoking* and *Be Moved* and *Be Moved* and *Be Moved* and *For Stations beyond Queen's Park take the first train* and *change at Queen's Park if necessary* and *Hammersmith* and *City Line* and *You're closer than my family ever gets*. And there were lots of people standing in the little station and it was underground so there weren't any windows and I didn't like that, so I found a seat which was a bench and I sat at the end of the bench.

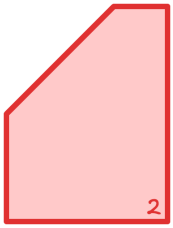
So I thought back to the origami puzzles and I looked at a new set of puzzles:



and I solved them easily:

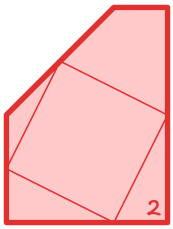


Then, I got to



and I was stuck until I made the *critical observation* that the diagonal edge can only mesh with itself nicely by being bisected.

Not soon after, I found the solution, and I thought it was very surprising:



Lots of people started coming into the little station, and someone sat down on the other end of the bench and I felt like I felt like when I had flu and I had to stay in bed all day and all of me hurt and I couldn't walk or eat or go to sleep or do maths.

I kept my eyes closed. The trains coming in and out of the station were in a rhythm, like music or drumming. I liked it. And I didn't open my eyes and I didn't look at my watch, and it was like being in a dark room with the curtains closed so I couldn't see anything, like when you wake up at night and the only sounds you hear are the sounds inside your head. That made it better because it was like the little station wasn't there, outside my head, but I was in bed and I was safe.

The silences between the trains coming and going got longer and longer. I could hear that there were fewer people in the little station when the train wasn't there, so I opened my eyes and I looked at my watch and it said 2:07 PM. and I had been sitting on the bench for approximately 3 hours but it hadn't seemed like approximately 3 hours, except that now my bottom hurt and I was hungry and thirsty.

And then I realised that Toby was missing because he was not in my pocket.

So I stood up and I looked up and down the little station and in the doorways that went into tunnels but I couldn't see him anywhere. I looked down into the black lower-down bit where the rails were. I saw two mice and they were black because they were covered in dirt, and I liked that because I like mice. But they weren't Toby, so I carried on looking.

When I finally found Toby, and he was also in the lower-down bit where the rails were, and I knew he was Toby because he was white and he had a brown egg shape on his back so I thought about the best way to save Toby without risking my own life and so I ended up cutting out a piece of string with my Swiss Army Knife and tied the remaining pellet of rat food onto the string and I *held on* to one end of the string and launched the other end to lure Toby and I saw that he had become attracted to my food and was walking towards it and so I slightly pulled on the string and he had followed it all the way out of the tracks and up the ledge to me and so I took him to me and I took off the pellet from the string and threw the string into the trash can nearby and I fed him the pellet that I just took off.

And then I heard the roaring and I covered my ears. That was probably the train.

The roaring got louder. I saw the train coming out of the tunnel.

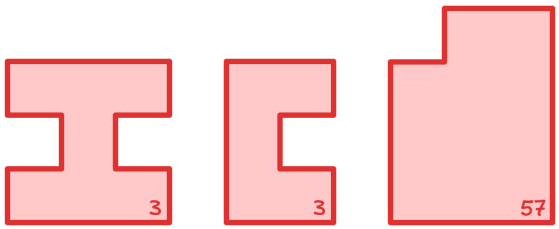
There was a sound like people fighting with swords and I could feel a strong wind and a roaring started and I closed my eyes and then the roaring turned into a clattering and a squealing and it got slowly quieter and then it stopped but I kept my eyes closed.

And then I could hear people moving again because it was quieter. And I opened my eyes but I couldn't see anything at first because there were too many people. When I finally could see, I went in.

The train doors closed and the train started moving and it roared again but not as loud this time and it went into the tunnel at the end of the little station with me.

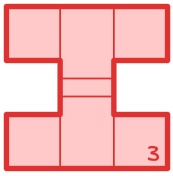
Toby tried to get out of my pocket again so I took hold of him and I put him in my outside pocket and I held him with my hand.

So I stayed there and did some more origami

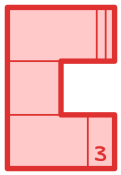


and I found those puzzles quite challenging.

I approached the first puzzle by first folding in the four extruded cells, and then realising that I had what I notate as a `313`, which meant three regions of thickness `3`, `1`, and `3` in a strip. The rest is simple: just tri-fold the `1` cell into a `3`.

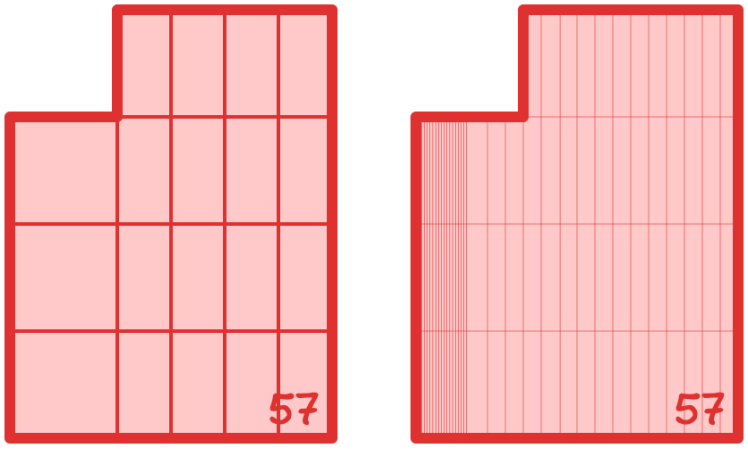


The second puzzle got me stuck for a while since I thought that I wanted to reduce from a `2 1 2`, but I soon realised that that was impossible and I had to work on the `3 2` from the other direction. Then I figured out that I can do this with a greedy approach, folding in just one of the two cells contributing to the `2` in half to create a `3 1` (with the `3` encompassing 1.5 blocks and `1` encompassing 0.5 blocks; that gets lost in the notation but is irrelevant in this case). `3 1` to `3` is trivial.



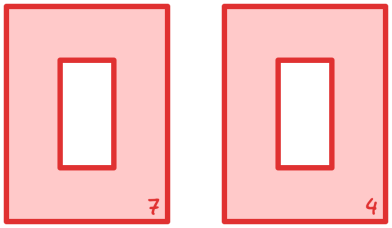
When the train stopped at Elephant & Castle on Baker Street and the doors opened automatically I still had not solved the third puzzle yet, so I walked out of the train. And then the doors closed and the train went away. And everyone who got off the train walked up a staircase and over a bridge except me, because I wanted to complete this puzzle.

I first realised that $57 = 3 \times 19$, and reducing column-wise gives `3 4` which already has a 3. Therefore, I just need to make everything else a 19, which meant I have to decompose 19 and $19 = 4 + 4 + 4 + 4 + 3$ works, so we fold the `4` in half twice and into the `3`, and finally fold the `3 19` into a `57`.



Chapter 179

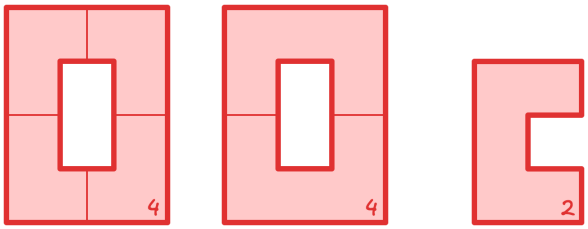
I then transferred to the Jubilee Line, which was the same drill as before so I won't bore you with the details. One thing I did was that I tried to do the next set of origami puzzles:



For thickness 7, we can fold the first and last rows inwards to get 212. We now want to get 7 from 2 and 1. We can have 1s left over, since we can simply fold them into a 7. Therefore, we use $7 = 2 + 2 + 2 + 1$. That is, we fold the 2s into 6s, fold them inward to get 717, and finally crimp 1 to get 7.

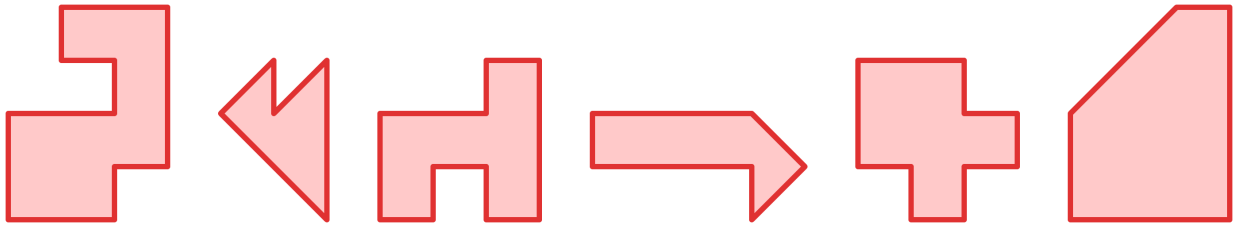
I tried to do the puzzle on the right in a similar way, but couldn't manage to do so. The figure reduces to 212 as well and so you can fold the 2s into 4s to get 414, but you cannot crimp 1 into 4 since it only works for odd thicknesses.

Therefore, I *played with it* some more.

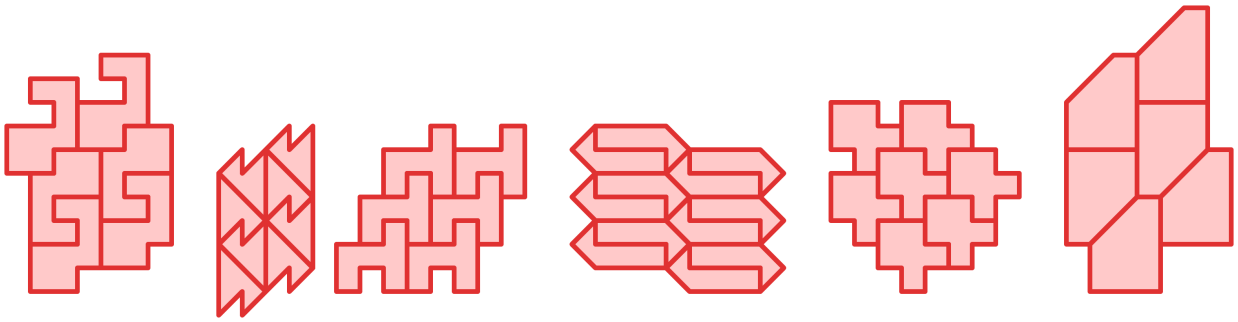


Since the shape was symmetrical, it made sense to fold it in half twice, but the end result is not a rectangle. So I backtracked to just one of the creases (the choice was quite natural, since that crease was on a grid line), which led me to a *reduction* of the problem, which meant that I only had to solve the C-shape on the right, but with thickness 2 instead of the thickness 3 from earlier in the puzzle set.

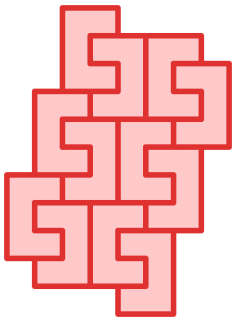
And then I was stuck for a while, so I looked back at the previous polygonal puzzles which had thickness 2, and I made a *critical observation*:



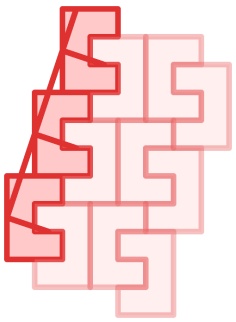
These shapes all tesselate, which means that they can tile the plane:



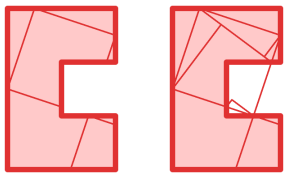
Then, I realised that the C-shape can also tesselate:



So if you take one strip, you can fold like this to *neutralise* the jagged edge:



The area of the shape is 5, and so the area of the rectangle is 5/2, so the rectangle is probably $\sqrt{10}/2$ by $\sqrt{10}/2$, which meant that I could complete the creases. So I drew out the creases and verified that it was correct indeed.



And so I had sent my friend this.

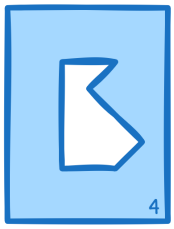
```
[14:35]Number Basher[THNK]
: p29 is col-wise 212 616 717 7
[14:37]Number Basher[THNK]
: p30 is red to 11/10/11 t2 tess > r10/2xr10/2 diag sol
[14:38]Number Basher[THNK]
: nice puzzs
```

Then, while I waited for a response I played some Tetris on `tetr.io` but only got to 900 meters because my reflexes are bad.

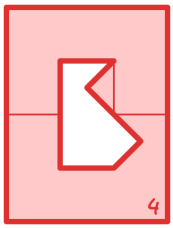
```
[14:46]Omega_3301[THNK]
: CHEEZ!
```

(A *cheese* is an alternate, unintended solution.)

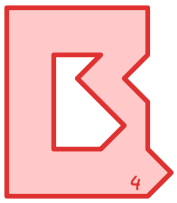
```
[14:48]Omega_3301[THNK]
:
```



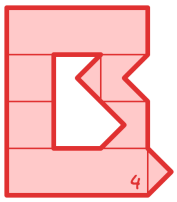

```
[14:48]Number Basher[THNK]
: uhhh
[14:50]Number Basher[THNK]
:
```



```
[14:51]Omega_3301[THNK]
: jkl;fdjaiowfejakdfl;haiwoe
[14:52]Omega_3301[THNK]
:
```



So, it was clear that we were supposed to reduce column-wise to `424` like this



but I still didn't know what was intended.

```
[15:20]NumberBasher[THNK]
: HOW!???!!?????
```

And I went off Jubilee Line on Dorris Hill and walked onto Chapter Road.

There it was.

451c, Chapter Road, London NW2 5NG.

I pressed the button on the front porch. No answer.

I waited.

It started to rain and I got wet and I started shivering because I was cold but I liked it.

So I took out my paperclip and decided to pick the lock on the front door, since it was my mother's front door.

And when I was picking the lock I heard this:

"I don't care whether you thought it was funny or not."

"Judy, look. I'm sorry, OK."

"Well, perhaps you should have thought about that before you made me look like a complete idiot."

"Christopher."

"What?"

"Christopher, Christopher, Christopher."

And I pushed her away because she was grabbing me and I didn't like it, and I pushed really hard and I fell over because an equal but opposite force was applied to me when I pushed her.

“What the hell is going on?”

“I’m so sorry, Christopher. I forgot.”

And I was lying on the ground and Mother held up her right hand and spread her fingers out in a fan so that I could touch her fingers, but then I saw that Toby had escaped out of my pockets so I had to catch him, and then I touched her fingers.

“I suppose this means Ed’s here.”

And there was a wall around the garden and Toby was climbing up the wall and out I grabbed him just fast enough and put him back in my pocket.

“He’s hungry. Have you got any food I can give him, and some water?”

“Where’s your father, Christopher?”

“I think he’s in Swindon.”

“Thank God for that.”

“But how did you get here?”

“I came myself.”

“Christopher, you’re soaking. Roger, don’t just stand there.”

“Oh my God. Christopher. I didn’t... Why are you here on your own?”

“Are you going to come in or are you going to stay out here all night?”

“Roger, please. Come on, Christopher, let’s go inside and get you dried off.”

“You follow Roger.”

“Go on or you’ll catch your death.”

That was a threat.

“I’ll run you a bath.”

Meanwhile, I walked round the flat to make a map of it in my head so I felt safer, and the flat was like this:



Chapter 181

And then Mother made me take my clothes off and get into the bath and she said I could use her towel, which was purple with green flowers on the end. And she gave Toby a saucer of water and some bran flakes and I let him run around the bathroom. And he did three little poos under the sink and I picked them up and flushed them down the toilet and then I got back into the bath again because it was warm and nice. Then Mother came into the bathroom and she sat on the toilet.

"Are you OK, Christopher?"

"I'm very tired."

"I know, love."

"You're very brave."

"Yes."

"You never wrote to me."

"Yes."

"Why, Christopher? Why didn't you write to me, Christopher? I wrote you all those letters. I kept thinking something dreadful had happened, or you'd moved away and I'd never find out where you were."

And I contemplated.

I should tell her the truth.

I should tell her the truth.

"Father..."

Should I?

And I contemplated.

I looked at her.

And I saw myself.

"Father... said you were dead. He said..."

And then it came from me, pouring out like the rain draining from the sky. I felt my emotion spill out and splatter all over the floor, submerging the room.

"He said you went into hospital because you had something wrong with your heart."

There's no more reason to hold back my emotion, so I let my emotion run wild like a horse on an empty field, and I let it trample over all the crops and run free.

"He... he said you had a heart attack and died and ..."

"And he kept all the letters in a cardboard box in his bedroom and I... I just found them..."

Some part of me felt free.

And I was scared but I was also relieved, almost like when Hiccup freed Toothless.

"What?"

"Oh my God."

"Oh, Christopher, I'm so sorry."

"It's fine. None of it was your fault."

"Bastard. The bastard."

"Christopher, let me hold your hand. Just for once. Just for me. I won't hold it hard."

And I didn't care any more since my feelings had already been set free, then.

"Let's get you out of the bath and dried off, OK?"

I got out of the bath and dried myself with the purple towel. But I didn't have any pajamas so I put on a white T-shirt even though it made me uncomfortable.

And while I was doing this Mother went into the kitchen and heated up some tomato soup because it was red.

And then I heard someone opening the door of the flat and there was a strange man's voice outside, so I locked the bathroom door.

"I need to speak to him."

"He's been through enough today already."

"I know. But I still need to speak to him."

And Mother knocked on the door and said a policeman wanted to talk to me.

"Are you Christopher Boone?"

"Yes."

"Your father says you've run away."

"Yes."

"Is this your Mother?"

"Yes."

"Why did you run away?"

"I wanted to find Mother."

"Do you want to go back to Swindon to your father or do you want to stay here?"

"I want to stay here."

"And how do you feel about that?"

"I want to stay here."

"Hang on. I'm asking your mother."

"He told Christopher I was dead."

"OK, let's... let's not get into an argument about who said what here. All I want to know is just whether —"

"Of course he can stay."

"Well, I think that settles it."

"Are you going to take me back to Swindon?"

"No."

"If your husband turns up and causes any trouble, just give us a ring. Otherwise, you're going to have to sort this out between yourselves."

And then the policeman went away and I had my tomato soup and Mr. Shears stacked up some boxes in the spare room so he could put a blowup mattress on the floor for me to sleep on, and I went to sleep.

And then I woke up and heard shouting and it was 2:31:45 AM.

"I'm talking to her whether you like it or not."

"Roger. Don't. Just —"

"I'm not being spoken to like that in my own home."

"I'll talk to you how I damn well like."

"You have no right to be here."

"No right? No right? He's my fucking son, in case you've forgotten."

"What in God's name did you think you were playing at, saying those things to him?"

"What was I playing at? You were the one that bloody left."

"So you decided to just wipe me out of his life altogether?"

"Now let us all just calm down here, shall we?"

"Well, isn't that what you wanted?"

"I wrote to him every week. Every week."

"Wrote to him? What the fuck use is writing to him?"

"Whoa, whoa, whoa."

"I cooked his meals. I cleaned his clothes. I looked after him every weekend."

"So you thought it was OK to tell him his mother was dead?"

"I looked after him when he was ill."

"Now is not the time."

"I worried myself sick every time he wandered off somewhere at night."

"Right, so that's why you just removed me from y'all lives."

"And you? What? You wrote him some fucking letters."

"So fucking be it!"

"Now is not the time!"

"You, butt out or I'll —"

"Ed, for God's sake —"

"I'm going to see him. And if you try to stop me —"

And then Father came into my room.

But I was holding my Swiss Army knife with the saw blade out in case he grabbed me.

"It's fine, Christopher. I won't let him do anything. You're all right."

"Christopher?"

"Christopher, I'm really, really sorry."

"About everything. About Wellington. About the letters. About making you run away."

"I never meant... I promise I will never do anything like that again."

"Shit. Christopher, please."

"Look, I was trying to protect you. I didn't know... if you knew... so..."

"I think you should go now."

And then the policeman came back because Mr. Shears had rung the police station and he told Father to calm down and he took him out of the flat.

"You go back to sleep now. Everything is going to be all right. I promise."

And when I was asleep I had one of my favorite dreams. Sometimes I have it during the day, but then it's a daydream. But I often have it at night as well.

And in the dream a nuclear explosion has caused everyone to be autistic just like me.

And we can associate together and I can go anywhere in the world and I know that no one is going to make me feel uncomfortable. But if I don't want to go anywhere I don't have to, and I can stay at home and eat broccoli and oranges and licorice laces all the time, or I can play computer games for a whole week, or I can just sit in the corner of the room and rub a £1 coin back and forward over the ripple shapes on the surface of the radiator.

And I go out of Father's house and I walk down the street, and it is very quiet even though and many people are out there but they are all spread out and we are all alone and

I can't hear any noise except birds singing and wind and the soothing sound of the shoes against the sidewalk, and if I stand very close to traffic lights I can ever hear the little click as the colours change.

And I take a ladder from Father's van and I climb up onto the roof, and when I get to the edge of the roof I put the ladder across the gap and I climb to the next roof.

And I stand in the surf and it comes up and goes over my shoes. And I don't go swimming in case there are sharks. And I stand and look at the horizon and I take out my long metal ruler and I hold it up against the line between the sea and the sky and I demonstrate that the line is a curve and the earth is round. And the way the surf comes up and over my shoes and then goes down again is in a rhythm, like music or drumming.

And then the dream is finished and I am happy.

Chapter 191

The next morning I had fried tomatoes for breakfast and a tin of green beans which Mother heated up in a saucepan.

"OK. He can stay for a few days."

"He can stay as long as he needs to stay."

"This flat is hardly big enough for two people, let alone three."

"He can understand what you're saying, you know."

"What's he going to do? There's no school for him to go to. We've both got jobs."

"Roger. That's enough."

"It's bloody ridiculous."

Then she made me some Red Zinger herbal tea with sugar in it but I didn't like it.

"You can stay for as long as you want to stay."

And after Mr. Shears had gone to work she made a telephone call to the office and took a compassionate leave. Then she said we had to go and buy some clothes for me to wear and some pajamas and a toothbrush and a flannel. Except there were too many people in John Lewis and I was already in a new place that I laid down on the floor next to the wristwatches and I screamed and Mother had to take me home in a taxi.

Then she had to go back to the shopping center to actually buy me some clothes and some pajamas and a toothbrush and a flannel, so I stayed in the spare room while she was gone and I thought more about the origami puzzle earlier and I had solved it.

And when Mother got home she brought me a glass of strawberry milk shake and showed me my new pajamas.

"I have to go back to Swindon."

"Christopher, you've only just got here."

"I have to go back because I have to sit my maths A level."

"You're doing maths A level?"

"Yes. I'm taking it on Wednesday and Thursday and Friday next week."

"God."

"The Reverend Peters is going to be the invigilator."

"I mean, that's really good."

"I'm going to get an A grade. And that's why I have to go back to Swindon."

"I don't know whether that's going to be possible."

"But I have to go."

"Let's talk about this some other time, OK?"

"OK."

"But I have to go to Swindon."

"Christopher, please."

And then I thought about proving $a + b = b + a$ on $\mathbb{N}$. Define $\mathbb{N}$ in first-order logic with the following Peano axioms:

- $0 \in \mathbb{N}$.
- $s : \mathbb{N} \to \mathbb{N}$.
- $n \in \mathbb{N} \implies s(n) \in \mathbb{N}$.
- $m, n \in \mathbb{N} \ \wedge \ s(m) = s(n) \implies m = n$.
- $n \in \mathbb{N} \implies \neg(s(n) = 0)$.
- $\phi(0) \ \wedge \ (\forall\, n \in \mathbb{N}\ \phi(n) \implies \phi(s(n))) \implies \forall\, n \in \mathbb{N}\ \phi(n)$.

Define addition as:

- $+ : \mathbb{N}^2 \to \mathbb{N}$.
- $a + 0 \mapsto a$.
- $a + s(b) \mapsto s(a + b)$.

We first prove that $a + 0 = 0 + a$ by induction on a:
Base Case: $a + 0 = 0 + 0 = 0 = 0 + 0 = 0 + a$.
Induction Hypothesis: $a + 0 = 0 + a$.
Induction Step: $s(a) + 0 = s(a) = s(a + 0) = s(0 + a) = 0 + s(a)$.

We now prove that $s(a) + b = s(a + b)$ by induction on b:
Base Case: $s(a) + 0 = s(a) = s(a + 0)$.
Induction Hypothesis: $s(a) + b = s(a + b)$.
Induction Step: $s(a) + s(b) = s(s(a) + b) = s(s(a + b)) = s(a + s(b))$.

Finally, we prove that $a + b = b + a$ by induction on b:
Base Case: $a + 0 = 0 + a$.
Induction Hypothesis: $a + b = b + a$.
Induction Step: $a + s(b) = s(a + b) = s(b + a) = s(b) + a$.

And hence we are done.

Chapter 193

But I couldn't sleep and I got out of bed and it was 12:34:56 AM. That was nice. And there was no one in the street and it was quieter than it was during the day, even though you could hear traffic in the distance and sirens. And I walked down Chapter Rd. and looked at all the cars and the patterns the phone wires made against the orange clouds and the things that people had in their front gardens, like a gnome and a cooker and a tiny pond and a teddy bear.

Then I heard two people coming along the road, so I crouched down between the end of a skip and a Ford Transit van, and they were talking in a language that wasn't English, but they didn't see me. And there were two tiny brass cogs in the dirty water in the gutter by my feet, like cogs from a windup watch.

And I liked it between the skip and the Ford Transit van so I stayed there for a long time. And I looked out at the street and on the floow I saw patterns of pentomino crosses. I wondered whether you could tessellate them, and it turns out that you could, and so I tried solving it as an origami puzzle and I found out that it could be solved for all positive integer thicknesses other than 1, and that there are actually two choices for the strip `t2`.

"Christopher...? Christopher...?"

Mother was running down the road, so I came out from between the skip and the Ford Transit van and she ran up to me.

"Jesus Christ,"

"If you ever do that again, I swear to God, Christopher, I love you, but..."

"I don't know what I'll do."

So she made me promise never to leave the flat on my own because it was dangerous and because you couldn't trust people in London because there were strangers. And the next day she had to go to the shops again and when she came back she brought some food pellets for Toby and three Star Trek videos and I watched them in the living room until Mr. Shears came home and then I went into the spare room again. And I wished that there was a garden but there wasn't.

And the day after that the office where Mother worked rang and told her she couldn't come back to work because they had got someone else to do her job for her, and she was really angry and she said that it was illegal and she was going to complain.

"Don't be a bloody fool. It was a temporary job, for Christ's sake."

"I have to go to Swindon to take my A level."

"Christopher, not now. I'm getting phone calls from your father threatening to take me to court. I'm getting it in the neck from Roger. It's not a good time."

"But I have to go because it's been arranged and the Reverend Peters is going to invigilate."

"Look. It's only an exam. I can ring the school. We can get it postponed. You can take it some other time."

"I can't take it another time. It's been arranged. And I've done lots of revision. And Mrs. Gascoyne said we could use a room at school."

"Christopher, I am just about holding this together. But I am this close to losing it, all right? So just give me some —"

Then she stopped talking and she put her hand over her mouth and she stood up and went out of the room. And I started feeling a pain in my chest because I thought I wasn't going to be able to go back to Swindon and take my A level.

And the next morning I looked out of the window in the dining room to look at the license plates on the cars in the street to do 4dx4d practice and I could just barely keep up with the cars, but if I concentrated on multiplying the license plate numbers it stopped me from thinking about my A level and the pain in my chest.

And in the afternoon Mother took me to Hampstead Heath in a taxi and we sat on the top of a hill and looked at the planes coming into Heathrow Airport in the distance. And I had a red ice lolly from an ice cream van.

And Mother said she had rung Mrs. Gascoyne and told her that I was going to take my maths A level next year, so I threw my red ice lolly away and I screamed for a long time and the pain in my chest hurt so much that it was hard to breathe.

"Is he OK?"

"Well, what does it look like to you?"

And then I was tired from screaming and Mother took me back to the flat in another taxi and the next morning was Saturday and she told Mr. Shears to go out and get me some books about science and maths from the library, and they were called *100 Number Puzzles* and *The Origins of the Universe and Nuclear Power*, but they were for children and they were not very good so I only pretended to read them.

"Well, it's nice to know my contribution is appreciated."

And when Mother and Mr. Shears talked I took the little radio from the kitchen and I went and sat in the spare room and I tuned it halfway between two stations so that all I could hear was white noise and I turned the volume up really loud and I held it against my ear so it overwhelmed my other senses and drowned away the hurt, like the hurt in my chest, and so then I couldn't hear Mother and Mr. Shears arguing and I couldn't think about not doing my A level or the fact that there wasn't a garden at 451c Chapter Road, London NW2 5NG, or the fact that I couldn't see where my future will lead.

And then it was Monday.

"You think you're so fucking clever, don't you."

I felt... what I think others would have called an overwhelming guilt, but what felt to me as simply pain.

"Don't you ever, ever think about other people for one second, eh?"

Excruciating pain.

"Well, I bet you're really pleased with yourself now, aren't you."

"Stop it!"

"Christopher, I'm sorry. I'm really, really sorry."

"Mother?"

"Why did you leave me?"

Chapter 197

The next morning, after Mr. Shears had gone to work, Mother packed lots of her clothes into two suitcases and told me to come downstairs and bring Toby and get into the car. And she put the suitcases into the boot and we drove off. But it was Mr. Shears's car.

"Are you stealing the car?"

"I'm just borrowing it."

"Where are we going?"

"We're going home."

"Home, as in Swindon?"

"Yes."

"Are we going back to Swindon so I can do my maths A level?"

"Please, Christopher. Don't give me any hassle right now, OK?"

"I'm meant to be doing my maths A level tomorrow."

"Just... Just... It's going to be all right, Christopher, OK? It's going to be all right."

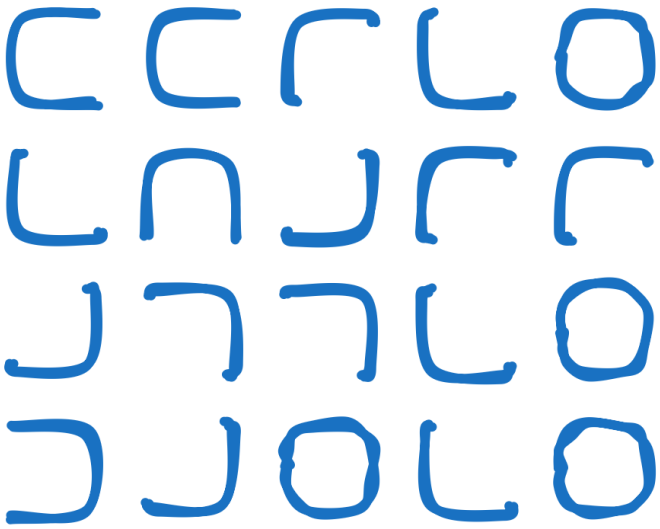
"We are going back to Swindon because if we stay in London any longer..."

"Someone was going to get hurt. And I don't necessarily mean you."

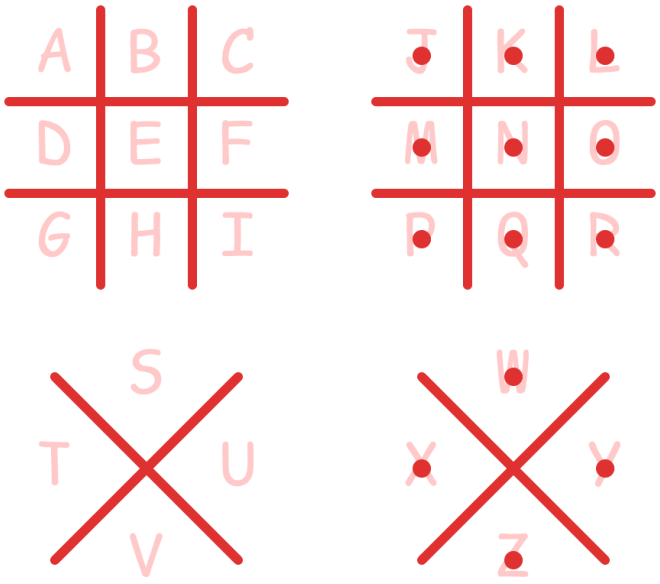
"What do you mean?"

"Now I need you to be quiet for a while."

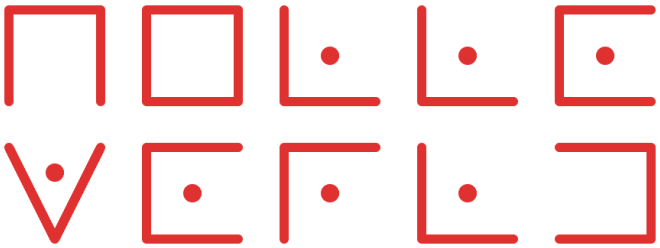
And since I had some time I had a friend send me a puzzle:



And this was called an *enigmatic-puzzle* since it doesn't tell you how to approach it. And so what I first saw was the three distinct shapes shaped like a C, L, and O in different rotations, and so I did a *Search* through my memory and remembered that there was this cipher called a pigpen cipher and this is how it worked:



So the letters get mapped to symbols, and hence `HELLO WORLD` gets translated to:



and so I thought that the puzzle could have been the pigpen cipher since many of the last eight letters were rare letters like `V`, `W`, `X`, `Y`, and `Z`.

So I decoded and got `FFICE`, `CHAII`, `AGGCE`, `DAECE`, which looked like real words (`OFFICE`, `CHAIR`, and `DANCE`). And then I noticed that there were no dots, and that I could add just one dot to make `CHAIR` and `DANCE`, and so I had made `FORCE`, `CHAIR`, `APPLE`, and `DANCE`.

And then I realised that they were all five-letter words and that the state of having or not having dots is a binary state, and so I extracted `01100`, `00001`, `01110`, and `00100` which were 12, 1, 14, and 4. Finally, we index these into the alphabet to get `LAND`.

And we drove all the way to Swindon and it took 3 hours and 12 minutes and we had to stop for petrol and Mother bought me a Milky Bar but I didn't eat it. And we got caught in a long traffic jam which was caused by people slowing down to look at an accident on the other carriageway.

So I tried to model the traffic jam, and I found out that you can expect yourself to be in a traffic jam that is worse than usual most of the time, since a worse traffic jam is weighted higher due to the longer time it takes. And so I knew that what I am experiencing is to be expected. And this effect is called survivorship bias since sampling is biased by the results.

Chapter 199

And when we got to Swindon Mother took out the keys and we went in.

"Hello?"

There was no one.

Mother said I would be safe, so I went up to my room and closed the door.

I took Toby out of my pocket and I let him run around and I played Minesweeper.

I did the Expert Version in 144 seconds, which was 46 seconds longer than my best time.

And then it was 6:35:24 PM.

I heard Father come home in his van and I moved the bed up against the door so he couldn't get in and he came into the house and he and Mother shouted at each other.

"How the fuck did you get in here?"

"This is my house, too, in case you've forgotten."

"Is your fucking fancy man here as well?"

And then I picked up the bongo drums that Uncle Terry had bought me and I knelt down in the corner of the room and I pressed my head into the join between the two walls and I banged the drums and I groaned and I carried on and then Mother came into the room and said Father had gone and then I banged some more and when that was done it had been precisely one hour and twelve minutes.

And I asked Mother if I could do my maths A level the next day.

"I'm sorry, Christopher."

Which was a no.

"I rang your headmistress. I told her you were in London. You'll do it next year."

"But I'm here now and I can take it."

"I'm sorry, Christopher. I was trying to do things properly."

"I was trying not to mess things up."

And I didn't like that so I groaned.

"I didn't know we'd be coming back."

But I carried on groaning and rocking backward and forward.

"Come on. This isn't going to solve anything."

Then she made me some carrots and broccoli and ketchup, but I didn't eat them.

And I didn't sleep that night either.

The next day Mother drove me to school in Mr. Shears's car because we missed the bus.

And when we were getting into the car, Mrs. Shears came across the road and said to Mother, "You've got a fucking nerve."

And Mother said, "Get into the car, Christopher."

But I couldn't get into the car because the door was locked.

"So, has he finally dumped you, too?"

Then Mother opened the door and unlocked my door and we got in and drove away.

"So you're Christopher's mother."

"Hi, Christopher, I'm glad to see you again."

"Are you still OK?"

"I am tired."

"He is upset since he couldn't do his maths A level."

And Siobhan said she would sort it out.

And after lunch Siobhan said that she had spoken to Mrs. Gascoyne and she still had my A level papers in 3 sealed envelopes in her desk. So I asked if I could still do my A level. And Siobhan said, "I think so. We're going to ring the Reverend Peters this afternoon to make sure he can still come in and be your invigilator. And Mrs. Gascoyne is going to write a letter to the examination board to say that you're going to take the exam after all. And hopefully they'll say that that's OK. But we can't know that for sure."

"I thought I should tell you now. So you could think about it."

"So I could think about what?"

"Is this what you want to do, Christopher?"

And I thought about the question and for the first time I wasn't sure what the answer was because I wanted to do my maths A level but I was very tired and when I tried to think about maths my brain didn't work properly and when I tried to remember certain facts, like the second taxicab number (which were numbers that can be expressed as the sum of cubes two different ways), I couldn't remember them and this made me frightened.

"You don't have to do it, Christopher."

"I want to do it."

"If you say you don't want to do it no one is going to be angry with you. And it won't be wrong or illegal or stupid."

"I want to do it."

I don't like it when I put things in my timetable and I have to take them out again.

"OK."

And she rang the Reverend Peters and he came into school at around 3:27 PM.

"So, young man, are we ready to roll?"

And I did Paper 1 of my maths A level sitting in the Art Room. And the Reverend Peters was the invigilator and he sat at a desk while I did the exam and he read a book called *The Curious Incident of the Dog in the Night-Time* and ate a sandwich. And in the middle of the exam he went out and smoked a cigarette, but he watched me through the window in case I cheated.

When I opened the paper and read through it, I didn't know how to answer any of the questions and also I couldn't breathe properly like there is something stuck in my nose. And I felt an urge to hit somebody or stab them with my Swiss Army knife, but that was wrong because people would feel sad. So I took deep breaths like Siobhan said I should and counted the cubes:

1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728, 2197, 2744, 3375, 4096, 4913.

And so I listed a table to find the second taxicab number:

a \ b	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1	2	9	28	65	126	217	344	513	730	1001	1332	**1729**	2198	2745	3376	4097	4914
2		16	35	72	133	224	351	520	737	1008	1339	1736	2205	2752	3383	**4104**	4921
3			54	91	152	243	370	539	756	1027	1358	1755	2224	2771	3402	4123	4940
4				128	189	280	407	576	793	1064	1395	1792	2261	2808	3439	4160	4977
5					250	341	468	637	854	1125	1456	1853	2322	2869	3500	4221	5038
6						432	559	728	945	1216	1547	1944	2413	2960	3591	4312	5129
7							686	855	1072	1343	1674	2071	2540	3087	3718	4439	5256
8								1024	1241	1512	1843	2240	2709	3256	3887	4608	5425
9									1458	**1729**	2060	2457	2926	3473	**4104**	4825	5642
10										2000	2331	2728	3197	3744	4375	5096	5913
11											2662	3059	3528	4075	4706	5427	6244
12												3456	3925	4472	5103	5824	6641
13													4394	4941	5572	6293	7110
14														5488	6119	6840	7657
15															6750	7471	8288
16																8192	9009
17																	9826

And so the second taxicab number was **4104**.

And that made me feel calmer. The exam was 2 hours long and 20 minutes had already gone by but it was fine since I can work really fast and I can check my previous answers while I am writing since writing isn't mental work.

And that night, just after I got home, Father came back to the house and I screamed but Mother said she wouldn't let anything bad happen to me and I went into the garden and laid down and looked at the stars in the sky and made myself negligible.

And the grass as wet.

And it was nice.

And the next day I did Paper 2 and the Reverend Peters finished *The Curious Incident of the Dog in the Night-Time*, but this time he didn't smoke a cigarette and Siobhan allowed me go into the toilets before the exam and sit on my own and do breathing and counting.

And I was playing *Tetris* on my computer that evening when a taxi stopped outside the house.

I was amused since I just brought up taxicab numbers.

Mr. Shears was in the taxi and he got out of the taxi and threw a big cardboard box of things belonging to Mother onto the lawn. Then he got some keys out of his pocket which were presumably car keys and got into his car and drove away and Mother ran out of the house and she ran into the street and shouted, "Don't fucking bother coming back, either!"

The next day I did Paper 3 and the Reverend Peters read *Diagnostic and Statistical Manual of Mental Disorders, 5th Edition* or just *DSM-5* and smoked three cigarettes. And they were all easy but I found this problem especially nice:

For a real number $\alpha > 0$, define $\alpha' = 1/(1 - 1/\alpha)$. Let $\mathcal{A}_\alpha = \{\lfloor k\alpha \rfloor \mid k \in \mathbb{N}_1\}$ and $\mathcal{B}_\alpha = \mathcal{A}_{\alpha'}$. Prove that $\mathcal{A}_\alpha$ and $\mathcal{B}_\alpha$ partition $\mathbb{N}_1$ iff α is irrational. And I was going to write out how I answered the question except Siobhan said it wasn't very interesting, but I said it was. And she said people wouldn't want to read the answers to a maths question in a book, so then I wanted to move everything before into an Appendix but Siobhan said I don't have to do that either. So to maintain consistency I'll leave this problem as an exercise to the reader.

But I still felt sick because I didn't know how I did in the exam because I didn't know if the examination board would allow my exam paper to be considered, but I calmed myself by telling myself that I can't change that which has already happened, which was true.

Chapter 211

Father came round to the house that night.

"Don't scream, OK, Christopher. I'm not going to hurt you."

Then he came a bit closer to me and he crouched down to show that he was not an Aggressor.

"OK."

"I wanted to ask you how the exam went."

"Tell him, Christopher."

"I don't know if I got all the questions right because I was really tired."

Silence.

"Thank you."

"What for?"

"Just... thank you."

"I'm very proud of you, Christopher. Very proud. I'm sure you did really well."

And then he went away.

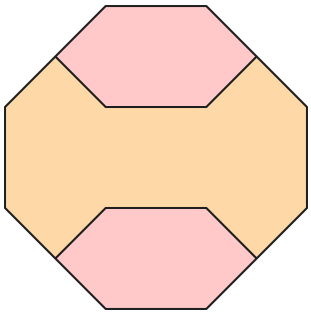
And the next week Father told Mother she had to move out of the house, but she couldn't because she didn't have any money to pay rent for a flat. But everything was fine because Mother got a job so we moved into a room in a big house, and the bed was in the same room as the kitchen and I didn't like it because it was small and the corridor was painted brown and there was a toilet and a bathroom that other people used and Mother had to clean it before I used it or I wouldn't use it and sometimes I wet myself because other people were in the bathroom.

And I didn't like waiting to find out about my maths A level. And whenever I thought about the future I couldn't see anything clearly in my head and that made a panic start. The future seemed to have such big range, with such uniform distribution, that it seemed hard to predict what will happen next. So Siobhan said I shouldn't think about the future.

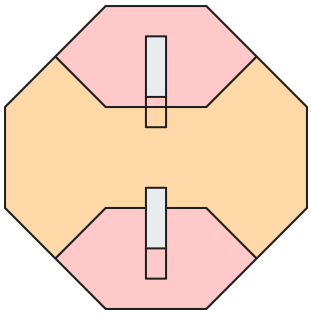
"Just think about today. Think about things that have happened."

But the things that happened were mostly bad things, and the worst thing was that Toby died because he was 2 years and 7 months old, which is very old for a rat, and I said I wanted to bury him, but Mother didn't have a garden, so I buried him in a big plastic pot of earth like a pot you put a plant in. And I said I wanted another rat but Mother said I couldn't have one because the room was too small.

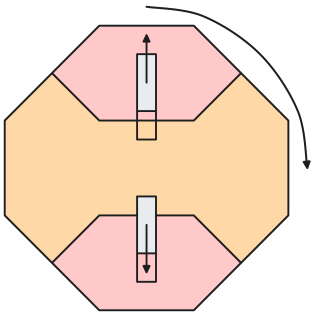
But there were good things, and one good thing was that I got a puzzle shaped like this:



And I solved the puzzle because I worked out that there were two bolts inside the puzzle and they were tunnels with metal rods in them like this



And normally you can't pull the pieces apart or have both rods slide to the outer halves, but you can if you use the typical trick of centrifugal force by spinning the shape:



"Christopher, can I have a talk with you? Five minutes, OK? That's all. Then you can go."

"Christopher, look... I'm really sorry. I don't know what I was thinking when I was killing Wellington. I... I might have thought that would make me more valued. But I don't know what I thought. And I'm sorry. And I know that with Mother, Wellington, now Toby... You have all the reasons to hate me. But things can't go on like this. I don't know about you, but this... it just hurts too much. You have to learn to trust me... and I will earn that trust, I will show that I deserve it. And I don't care how long it takes... If it's five minutes today and six minutes tomorrow and it takes years I don't care. Because this is important. This is more important than anything else."

"Sometimes, when this flawed world seems unusually hateful, I wonder whether there might be some other place, far away, where I should have been. I cannot seem to imagine what that place might be, and if I can't even imagine it then how can I believe it exists? And yet the universe is so very, very wide, and perhaps it might exist anyway? But the stars are so very, very far away. It would take a long, long time to get there, even if I knew the way. And I sometimes wonder what I would dream about, if I slept for a long, long time..."

And I completed the pattern.

"But no. Not Heaven, not some faraway star, not a different place but a better person, I'll show you, someday I'll show you how to be happy."

"We'll call it... we'll call it a project. A project we have to do together. You have to spend more time with me. And I... I have to show you that you can trust me. And it will be difficult at first because... because it's difficult. But it will get better. I promise."

And the alarm went.

Five fingers, spread out, like a fan.

And I hugged him.

And I really did mean it.

"You... you don't... you don't have to do this, not right now. I've got you a present. To show you that I really mean what I say. And to say sorry. I had thought... I had thought... when I killed... Wellington, that I would... I would get more... more... more love..."

"But... I was... I was wrong... Because... because... you have always... always... loved me, and I... I will... I will always... always... and I mean... always... love you..."

And then I felt a lick, and I saw that it was a golden retriever.

"He's... two months old. And he's... a golden... retriever."

And I let go, and I struck the dog. And no one said anything for a while.

"Christopher, I would never, ever do anything to hurt you."

"Does he have a name?"

"No. You can decide what to call him."

I called him Eddie.

One week letter, I got the results of my maths A level and I got an A grade, which is the best result and expected, so I was happy.

Father bought Eddie a collar and a lead and I was allowed to take him for walks to the shop and back, and I played with him with a rubber bone.

Deep inside, I know I can do anything. I seek into my future, and I see nothing concrete. I see an endless space of possibilities. And I may never know what step lies ahead, but *the big picture* is that everything will turn out well.

Wrap-up

At https://www.notion.so/numbas/Wrap-up-265a9dce9b0b80088910eebeb2b4362b .